Natural boundaries of the full square-lattice Ising susceptibility in two complex coupling variables

Spectral domains and physical-slice restrictions

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Abstract

We determine natural boundaries for the bulk zero-field susceptibility of the infinite square-lattice Ising model with two independent complex couplings. The high- and low-temperature physical germs extend holomorphically to explicit components of a spectral complement in $\mathbb{C}^2$. At resonant edges, a nonzero leading derivative survives the differentiated particle tail. At negative edges of the odd sum, exact pole periods give a single dominating particle index at selected Taylor orders. The regular-edge estimates apply along holomorphic transverse curves with nonramified edge coordinate and are locally uniform in complex dispersion parameters. An even–odd tail estimate treats the collinear isotropic restriction. Exact pullbacks give the natural boundaries at every fixed positive coupling ratio, and the map $s = \sinh z$ gives the isotropic disk and exterior germs.

1 Main results and the physical observables

The zero-field susceptibility sums spin correlations over the entire square lattice. Its form-factor expansion leads to a noncancellation problem for an infinite series of singular integrals [3, Section 3.2][4, Section 6]. With two independent couplings the analytic problem includes both the joint continuation and the continuations on the physical temperature lines.

The variables K_x, K_y are independent coupling coordinates. Put

$$z = 2K_x = x + \mathrm{i}y, \qquad w = 2K_y = u + \mathrm{i}v,$$
$$S = \sinh z, \quad T = \sinh w, \quad A = \cosh z \cosh w, \quad p = ST.$$

For real positive couplings, $\mathcal{X}_H$ is the sum of all two-spin correlations in the unique high-temperature Gibbs state; $\mathcal{X}_L$ is the sum of all connected two-spin correlations in the infinite-volume plus phase. Section 3 gives the physical definitions and identifies their form factors.

Define the spectral set by

$$\Sigma = \{(K_x, K_y) : A = Sa + Tb \text{ for some } a, b \in [-1, 1]\}.$$

The two domains used in this paper are explicitly

$$\mathcal{D}_H = \{|y| < \pi/2,\ |v| < \pi/2,\ \cosh x \tanh|u| < \cos y, \quad \cosh u \tanh|x| < \cos v\}, \tag{1.1}$$
$$\mathcal{D}_L = \{x > 0,\ u > 0, \quad \cosh x \tanh u > |\cos y|, \quad \cosh u \tanh x > |\cos v|\}. \tag{1.2}$$

In these coupling coordinates $\mathcal{D}_H$ is unbounded along the decoupled directions and its positive-ratio temperature slices are bounded.

Theorem 1.1 (Joint physical natural boundaries)**.** *The following assertions hold for both phases of the full bulk reduced susceptibility.*

1. *The sets $\mathcal{D}_H$ and $\mathcal{D}_L$ are precisely the components of $\mathbb{C}^2 \setminus \Sigma$ containing, respectively, the real positive regions $0 < p < 1$ and $p > 1$. They are contractible and pseudoconvex. On them $|p| < 1$ and $|p| > 1$, respectively.*

2. *The physical germs extend to single-valued jointly holomorphic functions $\mathcal{X}_H \in \mathcal{O}(\mathcal{D}_H)$ and $\mathcal{X}_L \in \mathcal{O}(\mathcal{D}_L)$. Their complete odd and connected-even form-factor sums converge normally on compact subsets, with every fixed mixed complex derivative.*

3. *At every point of the respective boundary, there is no holomorphic function on a neighborhood agreeing with that physical continuation on an accumulating connected component of its domain side. These are thus natural boundaries also for continuation in unramified Riemann domains over $\mathbb{C}^2$, reached from the given physical germ.*

4. *At a dense set on every regular boundary face, the complete physical sum is nonextendable along every holomorphic transverse curve on which the local edge parameter is nonramified. The estimates are locally uniform for a fixed selected phase and compact admissible subfamilies of bounded holomorphic complex-parameter families on common polydisks in a regular-edge chart. The collinear equal-coupling restriction has the dense local obstructions of Proposition 6.8.*

Here regular faces and their edge parameters are specified by Definition 2.8 and Proposition 2.10.

For every finite $\lambda > 0$, put $\iota_\lambda(\zeta) = (\zeta/2, \lambda\zeta/2)$. Let

$$P_\lambda(a) = \cosh a \tanh(\lambda a), \qquad Q_\lambda(a) = \cosh(\lambda a) \tanh a, \qquad Y_\lambda = \frac{\pi}{2\max(1,\lambda)}.$$

Both functions on $[0,\infty)$ have increasing inverses. Define

$$\begin{aligned} H_\lambda(b) &= \min\{P_\lambda^{-1}(\cos b), Q_\lambda^{-1}(\cos(\lambda b))\}, && |b| < Y_\lambda, \\ L_\lambda(b) &= \max\{P_\lambda^{-1}(|\cos b|), Q_\lambda^{-1}(|\cos(\lambda b)|)\}, && b \in \mathbb{R}, \\ \Omega_{H,\lambda} &= \{a + \mathrm{i}b : |b| < Y_\lambda,\ |a| < H_\lambda(b)\}, \\ \Omega_{L,\lambda} &= \{a + \mathrm{i}b : a > L_\lambda(b)\}. \end{aligned}$$

Theorem 1.2 (Exact recovery of every positive-ratio slice)**.** *For every finite positive λ,*

$$\iota_\lambda^{-1}(\mathcal{D}_H) = \Omega_{H,\lambda}, \qquad \iota_\lambda^{-1}(\mathcal{D}_L) = \Omega_{L,\lambda}.$$

The restrictions of the two joint physical functions are the respective physical reduced susceptibilities, holomorphic on these exact domains, and nonextendable through every boundary point in the componentwise sense of Theorem 1.1. They retain the complete lattice sum and every odd or nonzero even particle order. On a physical line with fixed $J_x > 0$, $J_y = \lambda J_x$, the same conclusions hold for the energy-field response

$$\chi(\zeta) = \frac{\zeta}{2J_x}\,\mathcal{X}(\iota_\lambda(\zeta)).$$

The proof of nonextension on a slice uses the transverse-curve estimates of Theorem 1.1 and the local isotropic estimates at equal coupling.

Corollary 1.3 (Exact isotropic descendants)**.** *Set $\lambda = 1$ and $s = \sinh\zeta$. The reduced HT physical germ is single valued and holomorphic on $|s| < 1$, with value one at zero; the reduced connected-plus-phase LT physical germ is single valued and holomorphic on $|s| > 1$. Every point of $|s| = 1$ is a natural-boundary point for each germ. Their complete normalized representations are*

$$\mathcal{X}_H(s) = \frac{(1-s^4)^{1/4}}{s} \sum_{N\geq 1,\ N \text{ odd}} C_N(1/s), \qquad \mathcal{X}_L(s) = (1-s^{-4})^{1/4} \sum_{N\geq 2,\ N \text{ even}} C_N(s),$$

where the coefficients are the full-site integrals in Section 6. The LT onsite term is $1-(1-s^{-4})^{1/4}$. The map sinh *is a biholomorphism from $\Omega_{H,1}$ to the disk and is the universal covering from $\Omega_{L,1}$ to the exterior, with deck translations $\zeta \mapsto \zeta + 2\pi\mathrm{i}k$. The energy-field response uses $\beta = \zeta/(2J)$ on the physical temperature continuation.*

1.1 Relation to earlier work

The unsquared anisotropic Nickel equation appears in [4, Section 6]. The natural-boundary question for the susceptibility involves the infinite particle sum on the physical complementary components of Σ. At regular faces Sections 4 and 5 control that sum using the first resonant derivative and the growth of boundary Taylor coefficients, respectively. Section 3.1 gives the real-temperature form-factor input.

The joint domains include points off the positive-ratio lines. For example, $z = \operatorname{arsinh}(1/2)$, $w = \mathrm{i}\pi/6$ lies in $\mathcal{D}_H$ and on no positive-ratio temperature line. At regular edges the independent coordinates $(C, B) = (A/S, T/S)$ allow both constant-B transversals and curves with varying B. The exact decoupled-axis restriction $\mathcal{X}_H(K_x, 0) = e^{2K_x}$ illustrates the distinction between joint and restricted continuation.

Section 2 determines the geometry of the physical domains. Section 3 constructs the holomorphic susceptibilities, and Sections 4–6 give the local obstructions. Section 7 proves the joint and slice theorems and the isotropic corollary.

AI Declaration. ChatGPT was used extensively in the mathematical research, proof development, and preparation of this manuscript.

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2 Independent complex spectral geometry

We determine the spectral complement before constructing the susceptibility. The calculation supplies the physical components, their local edge coordinates and their exact temperature slices. Its input is the spectral equation alone. Write

$$\begin{aligned} z = 2K_x = x + \mathrm{i}y, \qquad w = 2K_y = u + \mathrm{i}v, \\ S = \sinh z, \quad T = \sinh w, \quad A = \cosh z \cosh w, \quad p = ST, \end{aligned}$$

and define the closed set

$$\Sigma = \{(K_x, K_y) : A = Sa + Tb \text{ for some } (a, b) \in [-1, 1]^2\}. \tag{2.1}$$

The two real functions that describe this set are

$$F_1 = \cosh^2 u \cos^2 y - \cosh^2 x \sinh^2 u, \qquad F_2 = \cosh^2 x \cos^2 v - \cosh^2 u \sinh^2 x. \tag{2.2}$$

The use of z, w instead of K_x, K_y is just an invertible linear coordinate change; all domains below are domains in the original $\mathbb{C}^2$ coupling space.

Theorem 2.1 (Exact spectral set)**.** *For two independent complex couplings, including zero, opposite-sign, and collinear real-part configurations,*

$$\Sigma = \{F_1 F_2 \le 0\}. \tag{2.3}$$

Proof. Reduction and the noncollinear case. The independent reflections $z \mapsto -z$ and $w \mapsto -w$ preserve Σ, by changing the sign of the corresponding representing coefficient. They also preserve F_1, F_2. We may therefore first suppose $x, u > 0$. Set

$$\begin{aligned} h = \sinh x, \quad c = \cosh x, \quad H = \sinh u, \quad C_0 = \cosh u, \\ t = \cos y, \quad q = \sin y, \quad V = \cos v, \quad Q = \sin v. \end{aligned}$$

For

$$D = \Im(\overline{S}T), \qquad N_a = \Im(\overline{A}T), \qquad N_b = \Im(\overline{S}A),$$

direct multiplication gives

$$\begin{aligned} D &= hC_0tQ - cHqV, \\ N_a &= ctVQ - hHC_0q, \qquad N_b = hcHQ - C_0tqV. \end{aligned} \tag{2.4}$$

When $D \neq 0$, the unique real solution of $A = Sa + Tb$ is $a = N_a/D,\ b = N_b/D$.

For the sign analysis put

$$\begin{aligned} a_0 = h^2, \quad b_0 = H^2, \quad A_0 = 1 + a_0, \quad B_0 = 1 + b_0, \\ X = t^2, \quad Y = V^2, \quad X_0 = \frac{A_0 b_0}{B_0}, \quad Y_0 = \frac{B_0 a_0}{A_0}. \end{aligned}$$

Expanding (2.4) and cancelling the common cross terms gives the identities

$$\begin{aligned} F_1 &= B_0(X - X_0), & F_2 &= A_0(Y - Y_0), \\ N_a^2 - D^2 &= F_2 K, & K &= (B_0 - Y)X - b_0, \\ N_b^2 - D^2 &= -F_1 L, & L &= a_0 - (A_0 - X)Y. \end{aligned} \tag{2.5}$$

Both K and L vanish at (X_0, Y_0), increase with X, and decrease with Y. If F_1, F_2 have strictly opposite signs, both differences of squares in (2.5) are strictly negative. Consequently $D \neq 0$ and $|a|, |b| < 1$; the point belongs to Σ.

Suppose instead that F_1, F_2 have the same strict sign and $D \neq 0$. For $0 < Y < 1$, membership would impose

$$X \leq \frac{b_0}{B_0 - Y}, \quad X \geq A_0 - \frac{a_0}{Y} \quad \text{if } F_1, F_2 > 0,$$

and the reverse two inequalities if $F_1, F_2 < 0$. But

$$\frac{b_0}{B_0 - Y} - \left(A_0 - \frac{a_0}{Y}\right) = \frac{A_0(Y-1)(Y-Y_0)}{Y(B_0 - Y)}. \tag{2.6}$$

For the positive-sign case $Y > Y_0$, this expression is negative; for the negative-sign case $Y < Y_0$, it is positive. Each case contradicts the required inequalities. At $Y = 0$, $F_2 < 0$ and $L = a_0 > 0$, so the negative-sign case is impossible directly from (2.5). At $Y = 1$, $K = b_0(X - 1)$ and $L = X - 1$; either same-sign case requires $X = 1$, which gives $D = 0$.

It remains to exclude a same-strict-sign point with $D = 0$ from Σ. Membership would give $N_a = N_b = 0$. Since $F_1 F_2 \neq 0$, (2.5) implies $K = L = 0$. Here $Y > 0$, and (2.6) then implies $Y = Y_0$ or $Y = 1$. The first contradicts $F_2 \neq 0$, whereas the second gives $X = 1$. At $X = Y = 1$ the segment condition is

$$cC_0 \leq h + H.$$

As

$$(cC_0)^2 - (h + H)^2 = (hH - 1)^2,$$

this requires $hH = 1$, and hence $F_1 = F_2 = 0$, again a contradiction.

Equality faces. Endpoint roots give a direct description of the equality cases and will also provide the later edge coordinates. At a point with $F_1 = 0$, $S \neq 0$, and $\tanh(z/2)$ is finite and nonzero. The two endpoint roots

$$r_+ = e^w \tanh(z/2), \qquad r_- = e^{-w} \tanh(z/2) \tag{2.7}$$

satisfy

$$\frac{r_+ + r_+^{-1}}{2} = \frac{A - T}{S}, \qquad \frac{r_- + r_-^{-1}}{2} = \frac{A + T}{S}. \tag{2.8}$$

Moreover,

$$|\tanh(z/2)|^2 = \frac{\cosh x - \cos y}{\cosh x + \cos y}. \tag{2.9}$$

Thus $F_1 = 0$ means that at least one of r_+, r_- has modulus one: the respective equations are $\cos y = \cosh x \tanh u$ and $\cos y = -\cosh x \tanh u$. The corresponding right side of (2.8) belongs to $[-1, 1]$. This supplies a representing pair with $b = 1$ or $b = -1$. Interchanging z, w proves membership when $F_2 = 0$.

Vanishing real parts. The remaining cases follow directly from the real and imaginary parts of the spectral equation. Let $x = 0, u > 0$. If $\cos v \neq 0$, the real part of the equation $A = Sa + Tb$ forces

$$b = \frac{\cosh u}{\sinh u} \cos y.$$

Its necessary condition is $|\cos y| \leq \tanh u$. Under that condition $\sin y \neq 0$, and

$$a = -\frac{\cos y \sin v}{\sinh u \sin y}$$

has modulus at most one, so the condition is also sufficient. Indeed $\cos^2 y \le \sinh^2 u/\cosh^2 u \le \sinh^2 u/(\sinh^2 u + \sin^2 v)$. Here $F_2 = \cos^2 v > 0$, and the condition is precisely $F_1 \le 0$. If $\cos v = 0$, all terms are imaginary, and the choice

$$a = 0, \qquad b = \tanh u \cos y$$

always represents A; here $F_2 = 0$. The case $u = 0, x > 0$ follows by exchange. Finally, if $x = u = 0$, the coefficients S, T are imaginary and $A = \cos y \cos v$ is real. Membership is equivalent to $A = 0$, represented by $a = b = 0$, and is exactly $F_1 F_2 = 0$. The initial reflections now cover all real-part signs. □

2.1 The physical components and their branches

The sign calculation now becomes a description of connected domains. The contractions used in the proof also fix the branches of the physical fourth-root factors.

Theorem 2.2 (All components; explicit physical domains)**.** *The components of* $\mathbb{C}^2 \setminus \Sigma$ *with* $F_1, F_2 > 0$ *are indexed by* $m, n \in \mathbb{Z}$ *and are*

$$F_1 > 0, \quad F_2 > 0, \quad |y - m\pi| < \pi/2, \quad |v - n\pi| < \pi/2.$$

Each is star-shaped about the point $(z, w) = (\mathrm{i}m\pi, \mathrm{i}n\pi)$*. The components with* $F_1, F_2 < 0$ *are the four sets specified by the independent signs of* x, u*; each is contractible.*

The component containing the origin is exactly

$$\mathcal{D}_H = \left\{ \begin{array}{l} |y| < \pi/2, \quad |v| < \pi/2, \\ \cosh x \tanh |u| < \cos y, \quad \cosh u \tanh |x| < \cos v \end{array} \right\}. \tag{2.10}$$

The component containing the positive real low-temperature region is

$$\mathcal{D}_L = \left\{ \begin{array}{l} x > 0, \quad u > 0, \\ \cosh x \tanh u > |\cos y|, \quad \cosh u \tanh x > |\cos v| \end{array} \right\}. \tag{2.11}$$

They satisfy the strict inequalities

$$|p| < 1 \text{ on } \mathcal{D}_H, \qquad |p| > 1 \text{ on } \mathcal{D}_L. \tag{2.12}$$

Proof. The sign description follows from Theorem 2.1. For $F_1, F_2 > 0$, both cosines are nonzero, so a connected component remains in a single indicated cosine strip in each variable. Within a fixed pair of strips, set $\eta = y - m\pi,\ \theta = v - n\pi$. The inequalities are

$$\cosh x \tanh |u| < \cos \eta, \qquad \cosh u \tanh |x| < \cos \theta.$$

Replacing (x, u, η, θ) by $t(x, u, \eta, \theta)$, $0 \le t \le 1$, decreases the left sides and increases the right sides. This proves star-shapedness and excludes further component subdivision.

For $F_1, F_2 < 0$, neither x nor u can vanish. Writing $a = |x|,\ b = |u|$, the conditions are

$$\cosh a \tanh b > |\cos y|, \qquad \cosh b \tanh a > |\cos v|.$$

Each left side is nondecreasing in both a, b, and strictly increasing in the variable inside its hyperbolic tangent. Choose once and for all $R > 0$ with $\sinh R > 1$. First increase (a, b) continuously to $(\max(a, R), \max(b, R))$. Then move these two coordinates linearly to (R, R), throughout staying in

$[R,\infty)^2$. During the second move both left sides are at least $\cosh R \tanh R = \sinh R > 1$. Finally move y, v linearly to zero. These three moves give a continuous contraction within the fixed sign component. The signs cannot change without leaving the set. This proves all component claims and the physical identifications (2.10)–(2.11).

The modulus identity, with the notation in the proof of Theorem 2.1, is

$$|p|^2 = (A_0 - X)(B_0 - Y). \tag{2.13}$$

For $F_1, F_2 > 0$,

$$A_0 - X < \frac{A_0}{B_0}, \qquad B_0 - Y < \frac{B_0}{A_0},$$

so $|p| < 1$. For $F_1, F_2 < 0$, both inequalities reverse, so $|p| > 1$. These arguments allow zero real parts as well. On the positive real quadrant the common sign of F_1, F_2 is the sign of $1 - (\sinh x \sinh u)^2$, which identifies the two real physical regions. □

Remark 2.3. In particular $\{|p| = 1\} \subset \Sigma$. The more specific corner identity is

$$A^2 - (\epsilon S + \delta T)^2 = (p - \epsilon\delta)^2, \qquad \epsilon, \delta \in \{1, -1\},$$

and therefore

$$\bigcup_{\epsilon,\delta\in\{1,-1\}} \{A = \epsilon S + \delta T\} = \{p^2 = 1\} \subset \Sigma.$$

The branches $(1 - p^2)^{1/4}$ on $\mathcal{D}_H$ and $(1 - p^{-2})^{1/4}$ on $\mathcal{D}_L$ are defined directly by the binomial series and their positive real physical values. No additional covering is needed for these factors. On $\mathcal{D}_L$, S, T are nonzero. On $\mathcal{D}_H$ the zeros of S and T are respectively $z = 0$ and $w = 0$; these are genuine complex subvarieties of the joint domain. Any particle representation used there must account for its removable normalization at these axes.

Lemma 2.4 (A holomorphic modulus description)**.** *Within the two central strips in* (2.10)*, the four inequalities defining $\mathcal{D}_H$ can equivalently be written*

$$|e^w \tanh(z/2)| < 1, \quad |e^{-w} \tanh(z/2)| < 1, \quad |e^z \tanh(w/2)| < 1, \quad |e^{-z} \tanh(w/2)| < 1.$$

The domain $\mathcal{D}_L$ is equivalently described by

$$\begin{aligned} &|e^{-w} \tanh(z/2)| < 1, \quad |e^{-w} \coth(z/2)| < 1, \\ &|e^{-z} \tanh(w/2)| < 1, \quad |e^{-z} \coth(w/2)| < 1. \end{aligned}$$

The functions in the second display are evaluated where they are finite; its inequalities themselves force $x, u > 0$.

Proof. For the first pair in the HT assertion, (2.9) gives

$$e^{2|u|} \frac{\cosh x - \cos y}{\cosh x + \cos y} < 1,$$

equivalent to $\cos y > \cosh x \tanh |u|$. Exchange variables for the other pair. In the first LT pair, multiplication forces $e^{-2u} < 1$, hence $u > 0$; the second pair similarly forces $x > 0$. If $r = |\tanh(z/2)|^2$, the first pair is $e^{-2u} < r < e^{2u}$, which is equivalent to $|\cos y| < \cosh x \tanh u$. Exchange variables again. □

Lemma 2.5 (Canonical decaying root)**.** *For $K = (K_x, K_y) \notin \Sigma$ and real q, the equation*

$$S\rho^2 - 2(A - T\cos q)\rho + S = 0 \tag{2.14}$$

has exactly one root of modulus less than one, interpreting that root as zero when $S = 0$. This root is holomorphic in the independent coupling variables and locally jointly holomorphic with the complex angle q. On a compact subset of $\mathbb{C}^2 \setminus \Sigma$, its modulus has a uniform gap below one for all real q. On both positive real physical components it is the positive decaying root.

Proof. When $S \neq 0$, put $W = (A - T\cos q)/S$. By the definition of Σ, $W \notin [-1, 1]$. The two roots are reciprocal, neither has modulus one, and exactly one lies in the unit disk. Equivalently, it is $\rho = W - \sqrt{W^2 - 1}$, with the square-root branch asymptotic to W at infinity. Its uniqueness makes these local holomorphic roots agree on overlaps. At $S = 0$, $A - T\cos q \neq 0$; the implicit function theorem at $\rho = 0$ supplies the same holomorphic root. Compactness gives both the modulus gap and a common thin complex angular neighborhood on any compact parameter set. For positive real S, T,

$$(A - S - T)(A + S + T) = (p - 1)^2 > 0$$

on either physical component, and hence $W > 1$. The interior root is then real and belongs to $(0, 1)$. □

2.2 Exact fixed-ratio pullbacks

A temperature line is obtained by substituting the fixed ratio into the joint inequalities. We perform this substitution exactly so that the later nonextension theorem has the correct one-variable domain.

Theorem 2.6 (Exact slice domains)**.** *For every finite $\lambda > 0$, let*

$$\iota_\lambda(\zeta) = (\zeta/2, \lambda\zeta/2), \qquad Y_\lambda = \frac{\pi}{2\max(1, \lambda)},$$

and, for $a \geq 0$, let

$$P_\lambda(a) = \cosh a \, \tanh(\lambda a), \qquad Q_\lambda(a) = \cosh(\lambda a) \, \tanh a.$$

Both functions are strictly increasing bijections of $[0, \infty)$ onto itself. Define

$$\begin{aligned} H_\lambda(b) &= \min\{P_\lambda^{-1}(\cos b),\ Q_\lambda^{-1}(\cos(\lambda b))\}, && |b| < Y_\lambda, \\ L_\lambda(b) &= \max\{P_\lambda^{-1}(|\cos b|),\ Q_\lambda^{-1}(|\cos(\lambda b)|)\}, && b \in \mathbb{R}. \end{aligned}$$

Then the exact equalities of open sets are

$$\iota_\lambda^{-1}(\mathcal{D}_H) = \Omega_{H,\lambda} = \{a + \mathrm{i}b : |b| < Y_\lambda,\ |a| < H_\lambda(b)\}, \tag{2.15}$$

$$\iota_\lambda^{-1}(\mathcal{D}_L) = \Omega_{L,\lambda} = \{a + \mathrm{i}b : a > L_\lambda(b)\}. \tag{2.16}$$

Both domains are connected and simply connected. There is a unique $z_c > 0$ with

$$\sinh z_c \sinh(\lambda z_c) = 1, \qquad P_\lambda(z_c) = Q_\lambda(z_c) = 1.$$

The function H_λ is positive inside its interval and extends continuously by zero at both endpoints, and $0 \leq L_\lambda \leq z_c$.

Proof. The derivatives

$$P'_\lambda(a) = \sinh a \tanh(\lambda a) + \lambda \cosh a \, \operatorname{sech}^2(\lambda a), \quad Q'_\lambda(a) = \lambda \sinh(\lambda a) \tanh a + \cosh(\lambda a) \operatorname{sech}^2 a$$

are positive on $[0, \infty)$, and both functions tend to infinity. On the slice, $(x, y, u, v) = (a, b, \lambda a, \lambda b)$. The two strip conditions in (2.10) are exactly $|b| < Y_\lambda$; its remaining two inequalities are $P_\lambda(|a|) < \cos b$ and $Q_\lambda(|a|) < \cos(\lambda b)$. This proves (2.15). In (2.11) the positive-real-part conditions are exactly $a > 0$, and the other inequalities invert to (2.16).

The positive function $\sinh a \sinh(\lambda a)$ increases strictly from zero to infinity. The identities

$$P_\lambda(a)^2 - 1 = \frac{\sinh^2 a \sinh^2(\lambda a) - 1}{\cosh^2(\lambda a)}, \quad Q_\lambda(a)^2 - 1 = \frac{\sinh^2 a \sinh^2(\lambda a) - 1}{\cosh^2 a}$$

prove the assertions about z_c and L_λ. At either endpoint of the HT interval at least one of its two cosine arguments is zero, which proves the endpoint limit for H_λ. The HT domain is star-shaped by Theorem 2.2, or directly by simultaneous scaling of a, b. The LT domain is homeomorphic to $(0, \infty) \times \mathbb{R}$ via $(a, b) \mapsto (a - L_\lambda(b), b)$. □

Lemma 2.7 (Isotropic covering geometry)**.** *At $\lambda = 1$,*

$$\iota_1^{-1}(\Sigma) = \{\zeta : |\sinh \zeta| = 1\},$$

$$\Omega_{H,1} = \{|b| < \pi/2, \ |a| < \operatorname{arsinh}(\cos b)\}, \qquad \Omega_{L,1} = \{a > \operatorname{arsinh}(|\cos b|)\}.$$

The map $s = \sinh \zeta$ maps $\Omega_{H,1}$ biholomorphically onto $\{|s| < 1\}$, and maps $\Omega_{L,1}$ as the universal covering of $\{|s| > 1\}$, with deck translations $\zeta \mapsto \zeta + 2\pi \mathrm{i} k$, $k \in \mathbb{Z}$.

Proof. Here $S = T = s$, $A = 1 + s^2$. For $s \neq 0$, the spectral condition is $s + s^{-1} \in [-2, 2]$, whose preimage is exactly the unit circle; $s = 0$ is not spectral. Also $P_1 = Q_1 = \sinh$, giving the displayed domains. On the central strip $|\Im \zeta| < \pi/2$, sinh is injective: the alternatives for equal hyperbolic sines are $\zeta_1 = \zeta_2 + 2\pi \mathrm{i} k$ or $\zeta_1 = -\zeta_2 + (2k + 1)\pi \mathrm{i}$, and only the first with $k = 0$ is possible there. The domain $\Omega_{H,1}$ is bounded, its boundary maps into $|s| = 1$, and its derivative has no zero inside. Its image is consequently both open and relatively closed in the disk, and contains zero; it is the whole disk.

For the exterior, put $\omega = e^\zeta$. The equation $\omega - \omega^{-1} = 2s$ has, for $|s| > 1$, exactly one root with $|\omega| > 1$. A root of modulus one would give $s \in \mathrm{i}[-1, 1]$. The outside root varies holomorphically, for example by using $\sqrt{s^2 + 1} = s\sqrt{1 + s^{-2}}$. Its domain is the continuous radial epigraph

$$\{ re^{\mathrm{i}b} : r > \exp(\operatorname{arsinh} |\cos b|) \}.$$

Exponentiation from the simply connected epigraph $\Omega_{L,1}$ is its universal covering, and the outside-root map is a biholomorphism to $|s| > 1$. The stated deck group follows. □

2.3 Regular boundary faces in two complex variables

The local picture is a parallelogram in the dispersion coordinate: one representing coefficient reaches an endpoint while the other remains free. The next two propositions first show that such faces are dense and then give coordinates in which one can cross them.

Definition 2.8. A point of $\partial \mathcal{D}_H$ or $\partial \mathcal{D}_L$ will be called a regular joint edge point if $S, T \neq 0$, $D = \Im(\overline{S}T) \neq 0$, exactly one of its unique representing coefficients a, b is 1 or -1, the other lies in $(-1, 1)$, and $p^2 \neq 1$.

Proposition 2.9 (Density including exceptional strata)**.** *Regular joint edge points are dense in each of $\partial\mathcal{D}_H$ and $\partial\mathcal{D}_L$. More precisely, the points on either boundary for which $x, u \neq 0$ and exactly one of F_1, F_2 vanishes are regular and are dense. Thus the excluded zero-real-part, collinear, and intersection strata contain no relatively open boundary patch.*

Proof. First suppose $x, u \neq 0$, $F_1 = 0$, $F_2 \neq 0$. The algebra (2.5), valid also for negative real parts because it only uses squares, reduces to

$$N_a^2 - D^2 = -\frac{\sinh^2 u}{\cosh^2 u} F_2^2 < 0, \qquad N_b^2 - D^2 = 0. \tag{2.17}$$

Thus $D \neq 0$, $|a| < 1$, and $|b| = 1$. Furthermore (2.13) gives

$$|p|^2 = 1 - \frac{F_2}{\cosh^2 u} \quad (F_1 = 0), \qquad |p|^2 = 1 - \frac{F_1}{\cosh^2 x} \quad (F_2 = 0). \tag{2.18}$$

At an HT face the other F_i is positive, and at an LT face it is negative. Consequently $p^2 \neq 1$. The case $F_2 = 0, F_1 \neq 0$ follows by exchange.

Here is an explicit approximation argument covering all remaining boundary configurations. For $a, b \geq 0$, set

$$r_1(a, b) = \cosh a \tanh b, \qquad r_2(a, b) = \cosh b \tanh a.$$

Both $r_i < 1$, both equal one, or both exceed one according as $\sinh a \sinh b < 1$, $= 1$, or > 1. The closure of $\mathcal{D}_H$ is

$$|y|, |v| \leq \pi/2, \qquad r_1(|x|, |u|) \leq \cos y, \qquad r_2(|x|, |u|) \leq \cos v. \tag{2.19}$$

Indeed, contracting all four coordinates toward zero strictly opens every active inequality, proving the reverse inclusion in this closure description. If $\sinh|x| \sinh|u| < 1$, the angular fiber of (2.19) is the closed rectangle

$$|y| \leq \arccos r_1, \qquad |v| \leq \arccos r_2.$$

If the product is one, both intervals collapse to zero. Every boundary point has at least one active rectangular face. Approximate its real-part magnitudes by $a_j, b_j > 0$ with $\sinh a_j \sinh b_j < 1$. For an active first face choose $|y_j| = \arccos r_1(a_j, b_j)$, with the sign giving $y_j \to y$, and choose v_j strictly inside the second interval with $v_j \to v$. If the first interval has collapsed at the limit, either sign is available. Preserve the original signs of nonzero real parts; choose either sign at a zero real part. This gives $x_j, u_j \neq 0$, $F_1 = 0, F_2 > 0$, tending to the given point. Use the other face if it is the active one. This also covers corners, strip endpoints, and the collapsed fiber at real product one.

For $\mathcal{D}_L$, the closure is exactly

$$x, u \geq 0, \qquad r_1(x, u) \geq |\cos y|, \qquad r_2(x, u) \geq |\cos v|. \tag{2.20}$$

Increasing both real parts by any positive amount opens all its inequalities and proves the closure assertion. A boundary point must have $\sinh x \sinh u \leq 1$: if the product exceeds one, both $r_i > 1$ and the point is interior. It must also have at least one active cosine inequality. For example, $x = 0$ forces the second inequality to be active, while $u = 0$ forces the first. Again choose $x_j, u_j > 0$ with product less than one, tending to x, u. At an active first face choose

$$|\cos y_j| = r_1(x_j, u_j), \quad y_j \to y, \qquad |\cos v_j| < r_2(x_j, u_j), \quad v_j \to v.$$

These choices are always possible in the appropriate local cosine intervals. When $r_i \to 0$, these intervals shrink to a cosine zero; when $r_i \to 1$, their endpoints approach a multiple of π. The strict choice in the second interval remains possible since $r_2(x_j, u_j) > 0$. This gives $F_1 = 0, F_2 < 0$, tending to the prescribed point. The other active face is treated by exchange. □

Proposition 2.10 (Holomorphic coordinates on a regular face)**.** *Where $S \neq 0$, set*

$$C = A/S, \qquad B = T/S.$$

Then

$$\det \frac{\partial(C,B)}{\partial(z,w)} = \frac{p^2-1}{S^3}. \tag{2.21}$$

At a regular joint edge, (C,B) are therefore local holomorphic coordinates and $\Im B \neq 0$. If the endpoint is $b = \epsilon$, use $h = C - \epsilon B$: locally the boundary is $\Im h = 0$, with $\Re h \in (-1,1)$, and the physical domain occupies exactly one of its two sides. If the endpoint is $a = \epsilon$, exchange the transfer axes for the same conclusion.

On each regular boundary patch, points at which its free coefficient is

$$\cos(2\pi k/q), \qquad q \text{ an odd prime}, \quad 1 \le k \le (q-1)/2,$$

are dense.

Proof. The four derivatives are

$$C_z = -\frac{\cosh w}{S^2}, \quad C_w = \frac{\cosh z\, T}{S}, \quad B_z = -\frac{T\cosh z}{S^2}, \quad B_w = \frac{\cosh w}{S},$$

which give (2.21). Also $D = |S|^2 \Im B$. In these coordinates the spectral condition is simply

$$C \in [-1,1] + B[-1,1].$$

For nonreal B, this is a closed parallelogram in the C-plane. Near an interior point of its edge with $b = \epsilon$, the condition for the corresponding representing coefficient is

$$b - \epsilon = \frac{\Im(C - \epsilon B)}{\Im B};$$

the other coefficient remains strictly between its endpoints. Hence one side is spectral and the other is the unique local complementary side. Its intersection with the physical component is that whole side after shrinking the neighborhood. Holding B fixed permits h to range over an open real interval on the edge. The stated prime cosine values are dense in $(-1,1)$, since for arbitrarily large odd primes the angles $2\pi k/q$ approximate every angle in $(0,\pi)$. The coordinate description then proves the density assertion. □

Remark 2.11. On an HT regular face $F_1 = 0$, the endpoint is $b = \operatorname{sign} u$; on an HT regular face $F_2 = 0$, it is $a = \operatorname{sign} x$. This follows from (2.8), since both cosines are positive in the physical strips. On an LT regular face $F_1 = 0$, the endpoint is $b = \operatorname{sign}(\cos y)$; on $F_2 = 0$, it is $a = \operatorname{sign}(\cos v)$. In particular both signs of endpoint must be treated in any proof for the full physical boundaries.

2.4 Regular arcs on each fixed-ratio slice

To prove the slice theorem we need dense regular points on each specified temperature line. The following argument establishes that one-variable density for every $\lambda \neq 1$, including the local finiteness of its exceptional points.

Proposition 2.12 (Every anisotropic slice has dense regular arcs)**.** *Fix $\lambda > 0, \lambda \neq 1$, and restrict S, T, A, p, D to $(z, w) = (\zeta, \lambda\zeta)$. On each boundary $\partial\Omega_{H,\lambda}$ and $\partial\Omega_{L,\lambda}$, all but a locally finite set of points have $D \neq 0$, exactly one endpoint coefficient, and a nonzero derivative of the corresponding meromorphic edge map*

$$h_\epsilon(\zeta) = \frac{A - \epsilon S}{T} \quad (a = \epsilon), \qquad g_\epsilon(\zeta) = \frac{A - \epsilon T}{S} \quad (b = \epsilon). \tag{2.22}$$

Such points lie on real analytic boundary arcs, and are dense on both full boundaries. On each such arc, the prime cosine values in Proposition 2.10 are dense in its free coefficient.

Proof. We first prove the required local finiteness of the collinear candidates. Off the imaginary axis, both real parts $x, \lambda x$ are nonzero. The strict opposite-sign case in (2.5) gives $D \neq 0$; the strict face identity (2.17) excludes $D = 0$ on a single zero face as well. Consequently a collinear candidate off the imaginary axis has $F_1 = F_2 = 0$, hence $|p| = 1$. Conversely $|p| = 1$ implies membership in Σ, by (2.12) applied to all complementary components. Thus

$$\{\, \zeta \in \iota_\lambda^{-1}(\Sigma) : \Re\zeta \neq 0,\ D = 0 \,\} = \{\, \zeta : \Re\zeta \neq 0,\ D = 0,\ |p| = 1 \,\}. \tag{2.23}$$

For $x > 0$, the modulus equation is

$$(\sinh^2 x + \sin^2 y)(\sinh^2(\lambda x) + \sin^2(\lambda y)) = 1. \tag{2.24}$$

If $|\sin y \sin(\lambda y)| < 1$, its left side starts below one and strictly increases to infinity with x. Its positive x derivative at the root gives a unique positive real analytic solution $x = X(y)$. At a simultaneous zero of $\cos y, \cos(\lambda y)$, the root tends continuously to zero.

If there are no simultaneous cosine zeros, X is analytic on all of $\mathbb{R}$. Since X is even near zero, $X'(0) = 0$, and $X(0) = z_c$. Direct differentiation therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}y} D(X(y) + \mathrm{i}y)\Big|_{y=0} = \lambda \coth(\lambda z_c) - \coth z_c \neq 0. \tag{2.25}$$

For the last inequality, $t \coth t$ is strictly increasing on $t > 0$, because its derivative has the positive numerator $\sinh t \cosh t - t$. The analytic function $D(X(y) + \mathrm{i}y)$ is therefore nonzero identically and has locally isolated zeros.

Simultaneous cosine zeros occur exactly when $\lambda = r/s$ in lowest positive terms with r, s both odd; their locations are $y_j = (2j+1)\pi s/2$. Both X and $D(X(y) + \mathrm{i}y)$ have period πs. Each interval between successive y_j's is a translate of the interval containing zero, so (2.25) proves nonidentity on every such interval. It remains to exclude accumulation at its endpoints. Put $\zeta = \mathrm{i}y_j + \eta$, $\eta = x + \mathrm{i}\delta$. For signs $\sigma, \tau \in \{1, -1\}$,

$$S = \mathrm{i}\sigma \cosh \eta, \qquad T = \mathrm{i}\tau \cosh(\lambda\eta),$$

and direct Taylor expansion gives

$$D = \sigma\tau(\lambda^2 - 1)x\delta + O(|\eta|^4). \tag{2.26}$$

Equation (2.24), written analytically in x^2, δ^2, has linear part $(1 + \lambda^2)(x^2 - \delta^2)$. The implicit function theorem gives

$$X(y_j + \delta) = |\delta| + O(|\delta|^3).$$

Inserting this into (2.26) shows $D \neq 0$ for all sufficiently small nonzero δ, since $\lambda \neq 1$. There is no endpoint accumulation.

On the imaginary axis the candidate condition is $\cos y \cos(\lambda y) = 0$, a locally finite set. At a point where the cosine zeros are not simultaneous, $|p| = |\sin y \sin(\lambda y)| < 1$; continuity and (2.23) exclude nearby off-axis collinear candidates. The simultaneous case was just handled. Reflection $x \mapsto -x$, leaving y fixed, supplies the same conclusion on the other side. The complete set of collinear candidates is consequently locally finite in the ζ-plane.

The remaining exceptions are also locally finite. The zeros of S, T, and of $p^2 - 1$, are discrete. Each of the four functions in (2.22) is nonconstant meromorphic: at zero its leading term is $1/(\lambda\zeta)$ or $1/\zeta$. Its poles and critical points are locally finite. Let E_λ be the union of these sets and the collinear candidates.

At a boundary point outside E_λ, the unique representing coefficients cannot both be strict: otherwise continuity would place a whole neighborhood inside the spectral set, contrary to being approached by the complementary domain. They cannot both be endpoints, by the corner identity in Remark 2.3. Exactly one is an endpoint. Its edge map has nonzero derivative. For example, at $a = \epsilon$,

$$\Im h_\epsilon = \Im(S/T)(a - \epsilon)$$

for the unique real representing coefficients near the point; $\Im(S/T) \neq 0$. Hence the spectral set is locally one closed side of the nonsingular real analytic arc $\Im h_\epsilon = 0$, and the physical domain occupies the other side.

Neither physical boundary has an isolated point: the HT boundary is the two continuous graphs $a = \pm H_\lambda(b)$, completed at their endpoints, and the LT boundary is the continuous graph $a = L_\lambda(b)$. Removing the locally finite set E_λ therefore leaves a dense subset of both boundaries. The edge map is a local biholomorphism and maps each small regular arc to a real interval. Prime cosine density now proves the final assertion. □

Remark 2.13. The regular right boundary of the HT lens has positive endpoint, and its regular left boundary has negative endpoint, since x and $u = \lambda x$ have the same sign. On the LT boundary the endpoint signs are those in Remark 2.11, with $v = \lambda y$. In particular the HT lens has a regular negative-endpoint side at every unequal positive ratio. At $\lambda = 1$, the spectral curve is collinear along the entire isotropic boundary and Lemma 2.7 describes the slice geometry.

2.5 Exceptional boundary points and componentwise continuation

A dense set of obstructions will suffice once nearby points on the physical side can be joined inside a common local component. The collar lemma establishes precisely this connectivity, including at exceptional boundary points.

Lemma 2.14 (Uniform local collar)**.** *Let $D = \mathcal{D}_H$ or $\mathcal{D}_L$, and let $P \in \partial D$. For every open neighborhood U of P, there is an open neighborhood $V \subset U$ of P such that $D \cap V$ is contained in a single connected component of $D \cap U$. Consequently every component E of $D \cap U$ with $P \in \overline{E}$ contains $D \cap V$. The same assertion holds for every $\Omega_{H,\lambda}$ and $\Omega_{L,\lambda}$.*

Proof. It suffices to work in a small ball contained in U. Use real coordinates (x, u, y, v), and first consider $\mathcal{D}_H$. Its two nontrivial defining functions in the central strips are

$$g_1 = \cos y - \cosh x \tanh |u|, \qquad g_2 = \cos v - \cosh u \tanh |x|.$$

At $P = (x_0, u_0, y_0, v_0)$, use the fixed direction $\nu = -P$. For an active inequality $g_1(P) = 0$, its derivative in that direction is

$$y_0 \sin y_0 + x_0 \sinh x_0 \tanh |u_0| + |u_0| \cosh x_0 \operatorname{sech}^2 u_0 > 0. \tag{2.27}$$

All terms are nonnegative in the closed central strips. If the last term vanishes then $u_0 = 0$, and activity forces $|y_0| = \pi/2$, making the first term positive. If $u_0 \neq 0$, the last term is positive. Although $|u|$ is not differentiable at zero, when $u_0 = 0$ the chosen direction has zero u component. The directional derivative along ν is then well-defined and continuous in a neighborhood of P. The exchanged calculation applies to an active g_2. Thus there is a fixed small $t > 0$ and a neighborhood of P on which every active inequality increases uniformly along $Q + s\nu$, $0 \leq s \leq t$. Inactive inequalities remain positive after shrinking this neighborhood and t. The central strip conditions are preserved by the same translation for nearby points and sufficiently small t; at a strip endpoint it moves strictly inward. It follows that $P + t\nu \in D$ and, for all sufficiently near $Q \in D$, the entire segment $Q + [0, t]\nu$ lies in $D \cap U$.

For $\mathcal{D}_L$, use instead the direction $\nu = (1, 1, 0, 0)$ and the defining functions

$$g_1 = \cosh x \tanh u - |\cos y|, \qquad g_2 = \cosh u \tanh x - |\cos v|.$$

On $x, u \geq 0$, their derivatives in this direction are

$$\sinh x \tanh u + \cosh x \operatorname{sech}^2 u > 0, \qquad \sinh u \tanh x + \cosh u \operatorname{sech}^2 x > 0.$$

The same translation also opens $x, u > 0$. The absolute cosine terms are fixed during the translation, so they create no differentiability issue. This proves the same segment property.

In either case choose a small ball about $P + t\nu$ contained in $D \cap U$. Shrink V about P until every translated endpoint $Q + t\nu$, $Q \in D \cap V$, lies in that ball. Join any two such endpoints by a straight segment in the ball and use their translation segments to join the original points. Thus $D \cap V$ lies in one component of $D \cap U$. A component accumulating at P meets $D \cap V$ and hence is that component.

On a fixed-ratio slice the HT direction $-P$ preserves the slice; the same derivative proof applies. For LT use the real direction $(1, \lambda, 0, 0)$, or equivalently the horizontal translation $\zeta \mapsto \zeta + s$ of the epigraph $a > L_\lambda(b)$. The preceding ball-and-segment argument is then entirely within the complex line. □

Definition 2.15. For $f \in \mathcal{O}(D)$ and $P \in \partial D$, a local holomorphic continuation through P means a holomorphic function F on an open neighborhood U of P which agrees with f on some connected component E of $D \cap U$ satisfying $P \in \overline{E}$. The component E may be the one reached by the continuation path.

Corollary 2.16 (Propagation from dense genuine obstructions)**.** *Let D be any of the domains in Lemma 2.14, and let $f \in \mathcal{O}(D)$. Suppose a dense subset $G \subset \partial D$ consists of points at which no holomorphic function on a neighborhood can agree with f on the full intersection with D. Then f has no local holomorphic continuation through any point of ∂D, even in the componentwise sense of Definition 2.15.*

Proof. If F, U, E gave such a continuation at P, Lemma 2.14 would give $V \subset U$ with $D \cap V \subset E$, and therefore $F = f$ on $D \cap V$. Choose $Q \in G \cap V$ and a neighborhood of Q contained in V. Restricting F gives the forbidden continuation at Q. □

Remark 2.17. Propositions 2.9 and 2.12 give the dense families for Corollary 2.16. For the susceptibility the required local obstruction is that of the normalized full sum. On a fixed-ratio slice it is obtained from the integral along the corresponding curve $B = B(t)$ in the coordinates of Proposition 2.10.

2.6 Pseudoconvexity of the spectral components

Proposition 2.18 (An explicit plurisubharmonic exhaustion)**.** *Every component of* $\mathbb{C}^2 \setminus \Sigma$ *has a continuous plurisubharmonic exhaustion. In particular the explicit physical components above are pseudoconvex.*

Proof. Set

$$m(K) = \min_{(a,b)\in[-1,1]^2} |A - Sa - Tb|.$$

Compactness of the square makes m continuous, and its zero set is exactly Σ. On a complementary component, each $-\log|A - Sa - Tb|$ is pluriharmonic, and

$$-\log m(K) = \sup_{(a,b)\in[-1,1]^2} -\log|A - Sa - Tb|$$

is continuous and locally bounded above. The submean inequality on every complex line, or the usual locally bounded supremum criterion, shows that it is plurisubharmonic. Consequently

$$\Phi(K) = |K_x|^2 + |K_y|^2 + \max\{0, -\log m(K)\}$$

is continuous and plurisubharmonic. It tends to infinity at every finite boundary point, because $m \to 0$, and at infinity by its quadratic term. More explicitly, any sequence in a bounded sublevel set has a convergent subsequence with m bounded away from zero; its limit lies in the same complementary component. Thus its sublevel sets are relatively compact there. □

Remark 2.19. The exhaustion establishes pseudoconvexity of the complementary components. Nonextension of the physical susceptibility follows from the whole-sum obstructions and the local continuation argument.

3 The physical observable and joint holomorphy

Throughout this paper $z = 2K_x$ and $w = 2K_y$ are independent complex coordinates. Write

$$S = \sinh z, \qquad T = \sinh w, \qquad A = \cosh z \cosh w, \qquad p = ST.$$

For real positive K_x, K_y, write η for the dimensionless uniform field. The zero-field responses are responses are

$$\mathcal{X}_H(K_x, K_y) = \sum_{(m,l)\in\mathbb{Z}^2} \langle\sigma_{0,0}\sigma_{m,l}\rangle, \qquad 0 < p < 1, \tag{3.1}$$

$$\mathcal{X}_L(K_x, K_y) = \sum_{(m,l)\in\mathbb{Z}^2} \big(\langle\sigma_{0,0}\sigma_{m,l}\rangle_+ - M^2\big), \qquad p > 1, \tag{3.2}$$

$$M^2 = (1 - p^{-2})^{1/4}.$$

The high-temperature state is the unique Gibbs state. The subscript + means the infinite-volume plus state; the fourth root in the last line is positive. These definitions include the origin, both axes and all quadrants. The low-temperature onsite term is $1 - M^2$.

3.1 Real-temperature form factors and state identification

The real noncritical form factors come from the exact solution. The finite-cylinder formulas at arbitrary separation are [1, (35)–(39)]. The finite-lattice matrix-element factorization is proved in [2, Theorem 6.1], with the paramagnetic vacuum change in its Remark 6.2. In our convention put

$$B = T/S > 0, \quad C = A/S, \quad \cosh\gamma(q) = C - B\cos q, \quad \gamma(q) > 0.$$

Then the infinite-plane coefficient is

$$f_N(m,l) = \frac{B^{N^2/2}}{N!} \int_{\mathbb{T}^N} \prod_{i=1}^{N} \frac{e^{-|m|\gamma(q_i)+ilq_i}}{\sinh\gamma(q_i)} \prod_{i<j} \left\{ \frac{\sin((q_i-q_j)/2)}{\sinh((\gamma(q_i)+\gamma(q_j))/2)} \right\}^2 \prod_{i=1}^{N} \frac{\mathrm{d}q_i}{2\pi}. \tag{3.3}$$

With positive real square roots the two correlations are

$$\langle \sigma_0 \sigma_{m,l} \rangle = \xi_H \sum_{\substack{N \geq 1 \\ N \text{ odd}}} f_N(m,l), \qquad \xi_H = (1-p^2)^{1/4} p^{-1/2}, \tag{3.4}$$

$$\langle \sigma_0 \sigma_{m,l} \rangle_+ = \xi_L \left(1 + \sum_{\substack{N \geq 2 \\ N \text{ even}}} f_N(m,l) \right), \qquad \xi_L = (1-p^{-2})^{1/4} = M^2. \tag{3.5}$$

This is the normalization of [3, (1.8)–(1.10)]. In particular the anisotropy exponent in (3.3) is $N^2/2$.

Two identifications connect these formulas with the infinite-plane observable: the particle sum must pass through the cylinder limit, and the limiting correlations must belong to the specified phase. On a real noncritical compact set the dispersion has a uniform positive gap and is analytic in an angular strip. The finite-circumference corrections $\eta, \Lambda^{-1}, \log\xi_T$ in [1] tend uniformly to zero, by the exponentially convergent trapezoidal rule applied to their sum-minus-integral expressions. The following uniform estimates allow the particle index to grow with the circumference. If L is the circumference, the unnormalized Ramond and Neveu–Schwarz sums of a strip-analytic function differ by $O(Le^{-hL})$: their Fourier expansions select the multiples of L, with respective factors 1 and $(-1)^k$. Apply this to γ and to $\log(1 - e^{-\gamma(q)-\gamma(q')})$. The mixed double-grid difference of $\log\sinh((\gamma(q)+\gamma(q'))/2)$ in [1, (25)] gives

$$|\Lambda_L^{-1}| + \sup_q |\eta_L(q)| \leq CLe^{-hL}, \qquad |\log\xi_{T,L}| \leq CL^2 e^{-2hL}.$$

Thus at each fixed separation the entire correction is bounded by $C_{m,l}C^N$, uniformly in L, N. Terms with $N > L$ vanish on the discrete grid. For the real roots $0 < \rho(q) = e^{-\gamma(q)} \leq r < 1$,

$$\left| \frac{\rho(q) - \rho(q')}{1 - \rho(q)\rho(q')} \right| \leq r < 1.$$

The matching argument in Lemma 4.1 is an algebraic identity on discrete angular grids as well as on continuous circles. It gives a common bound $\exp\{-cN^2 + O(N\log(N+1))\}$ at fixed separation, including the finite-size one-body corrections and factorial normalization. Consequently the particle sum passes through the cylinder limit; each fixed-index Riemann sum gives (3.3).

Every local cylinder limit is a translation-invariant and spin-flip invariant Gibbs state. The Gibbs-state theorem [5, Theorem 1] applies with the vertex-transitive group chosen to be translations: the two unequal positive nearest-neighbor couplings satisfy its nonnegativity, invariance, irreducibility and summability hypotheses. That state is the equal mixture of the plus and minus states. Its two-spin correlations equal those of the plus state. This proves the state identification in (3.5); subtracting $M^2 = \xi_L$ removes exactly its zero-particle term. No response of a zero-mean mixed state is substituted for (3.2).

3.2 Double contours and the full lattice

The spatial geometric series of the double contour counts all lattice sites, and the two Schur products give summable bounds on compact coupling sets.

Let ζ, η now denote independent contour variables, and set

$$\mathcal{D}(\zeta,\eta;z,w) = A - \frac{S}{2}(\zeta+\zeta^{-1}) - \frac{T}{2}(\eta+\eta^{-1}), \qquad a(v,v') = \frac{v-v'}{1-vv'}.$$

If $X = \prod_i \zeta_i$ and $Y = \prod_i \eta_i$, the same coefficient has the symmetric representation

$$f_N(m,l) = \frac{p^{N/2}}{N!} \oint\cdots\oint X^{|m|-1} Y^{|l|-1} \prod_{i<j} a(\zeta_i,\zeta_j) a(\eta_i,\eta_j) \prod_{i=1}^{N} \frac{\mathrm{d}\zeta_i \, \mathrm{d}\eta_i}{(2\pi\mathrm{i})^2 \mathcal{D}(\zeta_i,\eta_i)}. \tag{3.6}$$

The circles have radii strictly less than and sufficiently close to one. On the real physical region, at $\eta = e^{\mathrm{i}q}$ the inside root is $\zeta = \rho = e^{-\gamma(q)}$, and

$$\mathcal{D}_\zeta(\rho,\eta) = \frac{S(1-\rho^2)}{2\rho^2}, \qquad \frac{\sqrt{p}}{\mathcal{D}_\zeta(\rho,\eta)} = \sqrt{B}\,\frac{\rho}{\sinh\gamma}.$$

Subtracting two dispersion equations gives the exact canceled pair

$$a(\rho_i,\rho_j)a(\eta_i,\eta_j) = -B\frac{(\eta_i-\eta_j)^2\rho_i\rho_j}{\eta_i\eta_j(1-\rho_i\rho_j)^2} = B\left\{\frac{\sin((q_i-q_j)/2)}{\sinh((\gamma_i+\gamma_j)/2)}\right\}^2 =: P_{ij}. \tag{3.7}$$

The residues therefore give $B^{N/2}$, and the pairs give $B^{N(N-1)/2}$. This proves the full exponent and the normalization in (3.3). Each spatial coordinate can be reflected independently at fixed couplings.

The exact product of the two spatial geometric series is

$$\sum_{m,l\in\mathbb{Z}} X^{|m|-1}Y^{|l|-1} = \frac{(1+X)(1+Y)}{XY(1-X)(1-Y)} = \frac{1}{XY} + \frac{2(X^{-1}+Y^{-1})}{(1-X)(1-Y)}. \tag{3.8}$$

The first term is the onsite coefficient. Denoting by T_N the integral with the second numerator but without its factor two gives $C_N = f_N(0,0) + 2T_N$, where C_N sums all sites. The angular form of this complete coefficient is

$$C_N = \frac{1}{N!}\int_{\sum_i q_i = 0 \pmod{2\pi}} G_N(q)\frac{1+Z}{1-Z}\prod_{i=1}^{N-1}\frac{\mathrm{d}q_i}{2\pi}, \qquad G_N = B^{N/2}\prod_i \frac{1}{\sinh\gamma_i}\prod_{i<j}P_{ij}, \quad Z = \prod_i \rho_i. \tag{3.9}$$

The $N-1$ independent normalized measures correspond to $q_1,\dots,q_{N-1}$, with $q_N = -\sum_{i<N} q_i$ modulo 2π.

Proposition 3.1 (Joint normal convergence)**.** *On each of the two domains $\mathcal{D}_H, \mathcal{D}_L$ specified by the independent spectral geometry, the physical germ has a single-valued jointly holomorphic continuation. It is given by*

$$\mathcal{X}_H = (1-p^2)^{1/4} \sum_{\substack{N\geq 1\\ N \text{ odd}}} p^{(N-1)/2} F_N, \tag{3.10}$$

$$\mathcal{X}_L = (1-p^{-2})^{1/4} \sum_{\substack{N\geq 2\\ N \text{ even}}} p^{N/2} F_N, \tag{3.11}$$

where F_N is the integral (3.6) *with its $p^{N/2}$ factor removed and with the multiplier* (3.8)*. The sums converge normally, with every fixed mixed complex derivative, on compact subsets. The fourth roots are the binomial branches supplied by $|p| < 1$ on $\mathcal{D}_H$ and $|p| > 1$ on $\mathcal{D}_L$.*

Proof. On a compact parameter neighborhood in $\mathbb{C}^2 \setminus \Sigma$, the polynomial $\mathcal{D}$ has a nonzero minimum modulus on the product of unit circles. By uniform continuity that gap persists on a common thin product annulus and a slightly larger parameter polydisk. Choose fixed radii $r_x, r_y < 1$ in the annulus. All Schur entries on these circles are bounded by a constant independent of N. For even N, the Schur Pfaffian identity and Hadamard's determinant inequality give

$$\left|\prod_{i<j} a(\zeta_i, \zeta_j)\right| = |\operatorname{Pf}(a(\zeta_i, \zeta_j))| \le C^{N/2} N^{N/4}.$$

For odd N, adjoin a ghost variable -1; its entries are the removable constant one and the dimension becomes $N+1$. Apply the estimate separately to the two contour-variable families. The one-body factors and contour lengths cost C^N. Also $1 - X$ and $1 - Y$ are bounded away from zero uniformly in N, whereas X^{-1}, Y^{-1} cost only C^N. Thus, after increasing constants,

$$|F_N| \le C_0 C_1^N \frac{N^{(N+1)/2}}{N!}. \tag{3.12}$$

The right side is summable after multiplication by any fixed exponential in N. Multivariate Cauchy estimates on the larger polydisk give the same assertion for every fixed mixed derivative. On overlaps, radii sufficiently close to one can be deformed into one another inside the common gap annulus; all pair and product denominators stay nonzero. The local integrals therefore glue. All powers of p within (3.10)–(3.11) are integers. The binomial fourth roots introduce no monodromy, and the series prove joint holomorphy, including the HT axes $S = 0$ or $T = 0$.

On a real physical compact set the same contours in (3.6) additionally give $r_x^{N|m|} r_y^{N|l|}$. Their double geometric sum is bounded uniformly in $N \ge 1$. Thus the lattice and particle sums are absolutely interchangeable, and (3.8) gives exactly (3.1)–(3.2), with the stated zero-particle subtraction. Equality on a real open physical set determines the joint germ: iterating the one-variable identity theorem over a real rectangle proves the required uniqueness. □

Matching the angular and double-contour branches. The boundary calculations use angular variables. To identify their sum with the holomorphic function just constructed, we first specify the canonical transfer root. For real q, outside Σ the equation

$$S\rho^2 - 2(A - T\cos q)\rho + S = 0$$

has a unique root of modulus less than one. For $S \neq 0$ it is the Joukowski inside root

$$\rho = W - \sqrt{W^2 - 1}, \qquad W = (A - T\cos q)/S, \qquad \sqrt{W^2 - 1} \sim W \quad (W \longrightarrow \infty).$$

At $S = 0$ the implicit-function theorem gives its removable value zero, since $A - T\cos q \neq 0$. This construction is jointly holomorphic locally in the couplings and in a complex angular neighborhood. On positive real noncritical parameters, $(A - S - T)(A + S + T) = (p-1)^2$ shows $W(q) > 1$, so it is the physical decaying root. On a patch with $\rho \neq 0$, use the local logarithm $\gamma = -\log \rho$.

The normalized angular coefficients agree with their double-contour counterparts throughout the physical domain. Indeed $1/\sinh\gamma = 2\rho/(1-\rho^2)$ and the rational expression for P_{ij} in (3.7) contain no logarithm. After including the physical normalization, their one-body prefactors are

$$(1-p^2)^{1/4} \frac{p^{(N-1)/2}}{S^N} \quad (N \text{ odd}), \qquad (1-p^{-2})^{1/4} \frac{p^{N/2}}{S^N} \quad (N \text{ even}).$$

All powers here are integers. Hence the normalized angular coefficient is single valued and holomorphic wherever $S \neq 0$ in the corresponding physical domain; the real angular torus has uniform root and denominator gaps on compact subsets. The set $\mathcal{D}_H \setminus \{S = 0\}$ is connected: radial contraction enters a small polydisk without crossing $z = 0$, and that polydisk minus $z = 0$ is connected. The domain $\mathcal{D}_L$ already has $S \neq 0$. The identity theorem from the real physical rectangle therefore gives equality with the double-contour coefficients throughout these sets. Near a regular edge $S, T \neq 0$, this also fixes the compatible local odd square root $\sqrt{B} = \sqrt{p}/S$. Normal convergence of the identified series follows from Proposition 3.1.

Remark 3.2 (The decoupled-axis restriction)**.** On $K_y = 0$, the lattice splits into independent horizontal chains. For $t = \tanh K_x$ near zero its response is

$$1 + 2 \sum_{m \geq 1} t^m = \frac{1 + t}{1 - t} = e^{2K_x}.$$

This agrees with (3.10): $p = 0$, only $N = 1$ survives, and $F_1 = (A - S)^{-1} = e^{2K_x}$. The restriction is entire, including at boundary points of the joint domain on this line. The slice theorem uses the obstructions along the positive-ratio temperature lines themselves.

3.3 Energy-field normalization

For fixed real $J_x, J_y > 0$ and physical temperature $(K_x, K_y) = (\beta J_x, \beta J_y)$, a field of energy units enters the Hamiltonian as $-h \sum_v \sigma_v$, and $\eta = \beta h$. Finite-volume differentiation gives $\chi = \beta \mathcal{X}$. The same statement holds after taking the prescribed infinite-volume phase limit. To justify the latter without exchanging an uncontrolled field limit and infinite sum, use the GHS inequality for arbitrary nonnegative couplings and fields [6, Theorem 1.5]. Pinning removed spins by fields tending to $+\infty$ shows that plus-box covariances increase with the box. Their nonnegative sums increase to the finite sum proved above. GHS concavity gives, for the plus-box magnetization,

$$m_V(h) - m_V(0) \leq h\, m_V'(0) \leq \beta h\, \mathcal{X}, \qquad h \geq 0.$$

The plus-state limit yields $m'(0+) \leq \beta \mathcal{X}$. Conversely turn on h only on a finite set E. The resulting bounded local change of measure has derivative $\beta \sum_{v \in E} \mathrm{Cov}_{+,0}(\sigma_0, \sigma_v)$. Ferromagnetic monotonicity bounds its magnetization above by the uniform-field magnetization, with equality at $h = 0$. Increasing E proves the reverse inequality. Thus $m'(0+) = \beta \mathcal{X}$. Spin-flip symmetry gives the two-sided HT derivative; in LT the derivative is the plus-phase right derivative (or its minus-phase counterpart). This statement is made on temperature lines only.

4 Regular spectral edges and transverse curves

At a resonant edge, strict contraction controls the form-factor numerators, and the differentiated tail is bounded. The leading amplitude is evaluated with holomorphic anisotropy along the curve, as needed on the physical temperature slices. On a regular edge, exchange the two lattice axes if necessary. In the local coordinates $C = A/S,\ B = T/S$ of the geometry section, one edge is $C - B = b$, $b = \cos\phi \in (-1, 1)$. Choose the side for which $\Im B_0 > 0$; conjugation handles the other side. Let $B(u)$ be any holomorphic function near zero with $B(0) = B_0$, and set

$$W(q; u) = b + u + B(u)(1 - \cos q). \tag{4.1}$$

Both a constant-B complex transversal in $\mathbb{C}^2$ and each nonramified physical temperature-line transversal have this form.

At $u = 0$, the inside root lies in the closed lower half disk, with $\rho(0) = e^{-\mathrm{i}\phi}$. Compactness gives

$$\Im\rho(q) \le -\delta < 0, \qquad |\rho(q)| \le 1.$$

There is no $W = \pm 1$ on the real angular circle. The roots, their chosen logarithms, the one-body factors, and the canceled pairs (3.7) are analytic on one fixed complex neighborhood in (u, B, q). On $u = \mathrm{i}\varepsilon$, $\varepsilon > 0$,

$$\Re\gamma(q; \mathrm{i}\varepsilon) \ge c\{\varepsilon + \mathrm{dist}(q, 0)^2\}. \tag{4.2}$$

Indeed $\Im W = \varepsilon + \Im B(\mathrm{i}\varepsilon)(1 - \cos q)$ has this lower bound; the identity $\Im W = \sinh(\Re\gamma)\sin(\Im\gamma)$ on the compact branch then gives (4.2). This is the complement side of the parallelogram edge and hence the side of the bordering physical component.

Lemma 4.1 (Strict contraction on a complex polydisk)**.** *There are $\kappa > 0$, $C < \infty$, and a fixed complex parameter polydisk about this edge such that, for every fixed mixed derivative ∂^ν in the parameters,*

$$|\partial^\nu G_N(q)| \le C_\nu \exp\{-\kappa N^2 + CN\log(N+1)\} \tag{4.3}$$

for all real angles and every $N \ge 1$.

Proof. The identity

$$|1 - vv'|^2 - |v - v'|^2 = (1 - |v|^2)(1 - |v'|^2) + 4\Im v\,\Im v'$$

gives a strict bound $|a(\rho_i, \rho_j)| \le q_0 < 1$. The denominator is separated from zero. By continuity a slightly larger constant below one still works on a common *complex* parameter polydisk, even at parameters where individual root moduli exceed one. Every canceled pair P_{ij} remains bounded by C.

For even N, expand the Schur Pfaffian in the $\eta_i = e^{\mathrm{i}q_i}$ variables only. For each matching, combine its $N/2$ edges with the corresponding ρ factors using (3.7). The other $N(N-2)/2$ root factors retain their strict contraction. Thus

$$\left|\prod_{i<j} P_{ij}\right| \le (N-1)!!\, C^{N/2} q_0^{N(N-2)/2}.$$

For odd N, adjoining the constant ghost entry gives

$$\left|\prod_{i<j} P_{ij}\right| \le N!!\, C^{(N-1)/2} q_0^{(N-1)^2/2}.$$

These identities first hold away from reciprocal η pairs and extend by the analytic canceled formula. There is no division by a root-Schur zero. One-body factors cost C^N. Cauchy's inequalities on the larger polydisk prove all the derivative bounds. They are applied to G_N, never across a pole of $(1+Z)/(1-Z)$. □

Lemma 4.2 (The entire differentiated higher-index tail)**.** *Suppose $\phi = 2\pi a/\ell$, where ℓ is an odd prime and $1 \le a < \ell/2$. For odd parity let $n = \ell$ and $k = (n^2-1)/2$; for even parity let $n = 2\ell$ and $k = n^2/2 - 1$. For every $0 \le j \le k$,*

$$\sum_{\substack{N > n \\ N \text{ of the chosen parity}}} |\partial_u^j C_N(\mathrm{i}\varepsilon)| = O_j(1) \qquad (\varepsilon \downarrow 0).$$

All lower indices of the chosen parity are holomorphic near $u = 0$.

Proof. Write $Q_0 = \sum_i \operatorname{dist}(q_i, 0)^2$. Equation (4.2) gives

$$|1 - Z| \geq c \min\{1, N\varepsilon + Q_0\}.$$

Since all fixed derivatives of γ are bounded, differentiating the exponential and rational function gives

$$\left|\partial_u^r \frac{1+Z}{1-Z}\right| \leq C_j N^j (1 + Q_0^{-j-1}), \qquad 0 \leq r \leq j.$$

If $N \geq 2j+4$, retain the first $2j+3$ independent angle squares in Q_0. Their inverse $(j+1)$-st power is integrable in that fixed real dimension; all unused normalized angular measures have mass one. Therefore Lemma 4.1 gives

$$|\partial_u^j C_N(\mathrm{i}\varepsilon)| \leq \frac{C_j N^j}{N!} \exp\{-\kappa N^2 + CN \log(N+1)\},$$

uniformly in ε.

For each remaining fixed N, the only configuration with $|Z| = 1$ is $q_1 = \cdots = q_N = 0$. There $Z = e^{-\mathrm{i}N\phi}$; the first resonant odd index is ℓ, and the first resonant even index is 2ℓ. Lower indices have a denominator gap and hence are holomorphic. Near the collision of a higher index use $q_N = -\sum_{i<N} q_i$, $Q = \sum_i q_i^2$, and $\Delta(q) = \prod_{i<j}(q_i - q_j)$. Every fixed numerator derivative has the factor Δ^2. The worst radial bound for the j-th derivative is

$$r^{N(N-1)} r^{N-2} (\varepsilon + r^2)^{-j-1} \, \mathrm{d}r.$$

Its limiting exponent $N^2 - 2j - 4$ is greater than -1 for every larger same-parity N and $j \leq k$. This handles the finite intervening range and completes the entire tail estimate. □

Proposition 4.3 (Nonzero full-sum edge obstruction on every holomorphic transversal)**.** *Under* (4.1) *and the rational-phase hypotheses of Lemma 4.2, the complete sum of the selected parity has no holomorphic extension through $u = 0$. The assertion is locally uniform in independent complex B parameters. It remains valid after multiplication by a holomorphic nonvanishing prefactor and after a nonramified change of transversal coordinate. It holds for both endpoint signs in the even case.*

Proof. The first amplitude. We evaluate the constrained Gaussian period with its exact measure. Its nonzero value supplies the leading coefficient that the bounded higher-index tail cannot cancel. Put $\alpha = (n^2 - 1)/2$ and

$$P_n = \prod_{r=1}^{n-1} \frac{r!}{2^r}.$$

The Gaussian Vandermonde period on the *independent-coordinate* measure is

$$\int_{q_n = -\sum_{i<n} q_i} e^{-Q} \Delta(q)^2 \, \mathrm{d}q_1 \cdots \mathrm{d}q_{n-1} = \frac{n! \, \pi^{(n-1)/2} P_n}{\sqrt{n}} =: \mathcal{G}_n. \tag{4.4}$$

To derive it, expand the ordinary n-variable Vandermonde determinant in monic Hermite polynomials. Their Gaussian norms are $\sqrt{\pi}\, r!/2^r$, so that integral is $n!\pi^{n/2} P_n$. Setting $x_i = q_i + m$, $\sum q_i = 0$, has Jacobian n, and the m integral is $\sqrt{\pi/n}$. Division proves (4.4).

On the unique collision chart,

$$\gamma(q; u) = \mathrm{i}\phi - \frac{\mathrm{i}}{\sin\phi} \left(u + \frac{B_0 q^2}{2} \right) + O(u^2 + uq^2 + q^4).$$

The leading singular integral for C_n is consequently

$$\frac{2\mathrm{i}^{1-n^2}B_0^{n^2/2}}{n!(2\pi)^{n-1}2^{n(n-1)}\sin^{n^2-1}\phi}\int\frac{\Delta(q)^2}{nu+(B_0/2)Q}\,\mathrm{d}q_1\cdots\mathrm{d}q_{n-1}. \tag{4.5}$$

For odd n the local square root of B is the one used in the physical angular coefficient. Its sign can change the displayed amplitude's sign but cannot make it zero. For even n the power in the original integrand is an integer; the square root below is fixed by the Gaussian period.

After k derivatives, the model integral is evaluated from

$$\int\frac{\Delta(q)^2}{(a+bQ)^{k+1}}\,\mathrm{d}q=\mathcal{G}_n\,b^{-\alpha}\frac{\Gamma(k+1-\alpha)}{\Gamma(k+1)}a^{\alpha-k-1},$$

initially in a common right half-plane and then by analytic continuation to $a=nu,\ b=B_0/2$. More explicitly, rotate both coefficients by $-\mathrm{i}$: $a'=n\varepsilon>0$ and $b'=-\mathrm{i}B_0/2$ has positive real part. The absolutely convergent Laplace–Gaussian calculation for a',b' fixes the continuation and its square root using $\Re b'>0$. For odd n, $k=\alpha$ is a multiple of four; for even n, $k=\alpha-\frac{1}{2}$ is odd. Substitution gives

$$\lim_{\varepsilon\downarrow0}u\,\partial_u^kC_n(u)=\frac{2^{1-k}\sqrt{B_0}\,P_nn^{k-3/2}}{\pi^{(n-1)/2}\sin^{2k}\phi}\neq0,\qquad u=\mathrm{i}\varepsilon,\quad n\text{ odd}, \tag{4.6}$$

$$\lim_{\varepsilon\downarrow0}u^{1/2}\partial_u^kC_n(u)=-\mathrm{i}\,\frac{2^{1-\alpha}\sqrt{B_0}\,P_nn^{k-1}}{\pi^{(n-2)/2}\sin^{n^2-1}\phi}\neq0,\qquad u=\mathrm{i}\varepsilon,\quad n\text{ even}. \tag{4.7}$$

In the second line $u^{1/2}=e^{\mathrm{i}\pi/4}\sqrt{\varepsilon}$.

Comparison with the exact integral. On the fixed-n collision chart write the exact integrand as $\Delta^2f(q,u)/H(q,u)$. It is even under simultaneous $q\mapsto-q$, and, with $c=-\mathrm{i}/\sin\phi$,

$$H=c\{nu+(B_0/2)Q\}+O((|u|+Q)^2),\quad H_u=cn+O(|u|+Q),\quad f=f_0+O(|u|+Q).$$

Both the exact and model denominators have modulus at least $c_1(\varepsilon+Q)$ on $u=\mathrm{i}\varepsilon$. The top derivative has leading term $(-1)^kk!fH_u^k/H^{k+1}$; all other terms have denominator power at most k. Subtracting the constant quadratic model therefore gives the pointwise bound

$$|\partial_u^k(\text{exact integrand}-\text{model integrand})|\le C\Delta^2(\varepsilon+Q)^{-k}.$$

Its radial integral is $O(\log(1/\varepsilon))$ for odd n, where $k=\alpha$, and $O(1)$ for even n, where $k=\alpha-1/2$. Extending the differentiated model from the chart to the whole tangent plane costs $O(1)$, since its radial tails are R^{-3} and R^{-2}, respectively. Outside the chart the exact denominator has a gap. The variation of $B(u)$ contributes $O(uQ)$ to the denominator remainder, so these estimates give both limits for the exact coefficient.

Passage to the full sum. Lemma 4.2 bounds all higher derivatives in the entire tail, and lower indices are analytic. Lower derivatives of the even first term are bounded; the next lower odd derivative may be logarithmic, but all are $o(u^{-1})$. Thus the nonzero top amplitudes cannot be canceled in the full sum or by product-rule terms from an analytic nonvanishing physical prefactor. The chain rule proves the statement for every nonramified curve parameter.

At the negative endpoint replace q_i by $\pi+q_i'$. For even N the total momentum constraint is unchanged and $B^{N^2/2}=(-B)^{N^2/2}$. Apply the preceding proof with $-B$, conjugating if necessary to have positive imaginary part. Exchanging transfer axes handles the other two edges. For odd N this shift instead makes the total reduced angle an odd multiple of π; that distinct case is addressed by the full boundary-jet theorem in the next section. □

5 The full odd sum at a regular negative spectral edge

At a negative edge, the odd momentum constraint prevents the all-endpoint configuration at any fixed particle number. Each inward derivative of the complete sum therefore has a finite boundary value. The obstruction appears in their growth: selected Taylor coefficients grow too rapidly to belong to a holomorphic function.

For a holomorphic anisotropy along a transverse curve, the proof uses the exact particle-dependent anisotropy factor and the full constrained angular integral. We reduce the coefficients to pole periods, evaluate the first two windings, and compare the dominating index with the other odd indices and windings.

5.1 Statement, branch, and the physical side

Fix

$$\phi = \frac{2\pi r}{p} \in (0,\pi), \qquad (r,p)=1, \qquad p \geq 3 \text{ odd}, \qquad b = \cos\phi, \qquad s_\phi = \sin\phi > 0. \tag{5.1}$$

Let B be holomorphic on a disk about zero and suppose that

$$\Im B(0) < 0. \tag{5.2}$$

Shrink this disk so that $B \neq 0$ and $\Im B < 0$ throughout it, and fix one holomorphic logarithm of B. For $q = \pi + \delta$, define

$$W(t,\pi+\delta) = b + B(t)\big(t - 1 + \cos\delta\big), \qquad g(t,\delta) = \operatorname{arcosh} W(t,\pi+\delta), \qquad g(0,0) = \mathrm{i}\phi. \tag{5.3}$$

At $t=0$, W is in the open upper half-plane except at $\delta = 0 \bmod 2\pi$, where it equals $b \in (-1,1)$. Consequently the branch in (5.3) extends to a fixed complex neighborhood of all real angles and a fixed parameter disk. It is even and 2π-periodic in δ. Put $\gamma(t,q) = g(t,q-\pi)$ and $\rho(t,q) = e^{-\gamma(t,q)}$. At $t=0$, $|\rho| \leq 1$, and the compact set of these roots lies strictly in the lower half-plane:

$$\Im\rho(0,q) \leq -\delta_0 < 0. \tag{5.4}$$

The logarithm used for γ is fixed by $0 < \Im\gamma(0,q) < \pi$. In particular all the one-body denominators and the canceled pair denominators below are nonzero on a fixed complex neighborhood.

For odd $n \geq 1$, let

$$\mathcal{T}_n = \{(q_1,\dots,q_n) \in (\mathbb{R}/2\pi\mathbb{Z})^n : q_1 + \cdots + q_n = 0 \bmod 2\pi\}, \qquad q_n = -\sum_{i<n} q_i \bmod 2\pi.$$

The measure on this torus is $\mathrm{d}q_1 \cdots \mathrm{d}q_{n-1}$. Include the normalization in the numerator:

$$H_n(t,q) = \frac{B(t)^{n^2/2}}{n!(2\pi)^{n-1}} \prod_{i=1}^{n} \frac{1}{\sinh\gamma(t,q_i)} \prod_{i<k} \left[\frac{\sin((q_i-q_k)/2)}{\sinh((\gamma(t,q_i)+\gamma(t,q_k))/2)}\right]^2, \tag{5.5}$$

$$\Gamma_n(t,q) = \sum_{i=1}^{n} \gamma(t,q_i), \tag{5.6}$$

$$C_n(t) = \int_{\mathcal{T}_n} H_n(t,q) \coth\!\left(\frac{\Gamma_n(t,q)}{2}\right) \mathrm{d}q. \tag{5.7}$$

The power of B uses the chosen common logarithm. Changing its square-root sign changes every odd coefficient by the same sign.

The relevant open side of the boundary is

$$\mathcal{U}^- = \{\, |t| < r_0 :\ \Im(B(t)t) > 0 \,\}. \tag{5.8}$$

Here

$$\Im W(t, \pi + \delta) = \Im(B(t)t) - \Im B(t)(1 - \cos\delta) > 0$$

for every real angle, so the chosen roots satisfy $|\rho| < 1$. This is the local physical side when the regular negative endpoint has the lower-half-plane root $\rho(0, \pi) = e^{-\mathrm{i}\phi}$. It contains $t = -\epsilon$ for every sufficiently small $\epsilon > 0$. When these coordinates are obtained from $W(q) = C - B\cos q$, they are exactly

$$t = \frac{C + B - b}{B}.$$

The curve hypothesis is that this t is nonramified, with $B(t)$ the anisotropy along the curve. Complex conjugation gives the opposite orientation.

Theorem 5.1 (Universal analytic-curve obstruction at a negative edge)**.** *Under* (5.1)–(5.2)*, the series*

$$F(t) = \sum_{\substack{n \geq 1 \\ n \text{ odd}}} C_n(t)$$

converges normally on $\mathcal{U}^-$. Every inward derivative has a finite limit

$$a_j = \lim_{\epsilon \downarrow 0} \frac{F^{(j)}(-\epsilon)}{j!}, \qquad j = 0, 1, 2, \ldots. \tag{5.9}$$

Nevertheless its formal boundary Taylor series has radius zero. More precisely, as n tends to infinity through odd multiples of p, there are integers j_n such that

$$j_n = n^2 \left(\log n + \tfrac{1}{2} \log\log n - c_* + o(1) \right), \tag{5.10}$$

$$|a_{j_n}|^{1/j_n} \sim \frac{4 j_n}{e\pi^2 \log j_n}, \tag{5.11}$$

$$c_* = -\frac{3}{4} - \frac{3}{2}\log 2 + \log\pi + \frac{1}{2}\log|B(0)| - \log s_\phi. \tag{5.12}$$

It follows that F has no holomorphic extension through $t = 0$ from $\mathcal{U}^-$. If P is holomorphic near zero and $P(0) \neq 0$, the same nonextension conclusion holds for PF.

The hypotheses and all estimates are locally uniform for complex holomorphic families $B(t, \eta)$: it suffices to work on a common parameter disk and a relatively compact parameter polydisk on which $\Im B(0, \eta) \leq -\beta < 0$ and B has a common holomorphic bound. The constants can depend on that disk, that bound, β, and the fixed rational phase. The selected integers may depend on the family parameter, $j_n = j_n(\eta)$.

We identify a single particle index that dominates all others in modulus at selected Taylor orders, allowing arbitrary phases for the other terms.

5.2 A uniform Gaussian numerator bound and all boundary jets

There are two estimates to establish first. Gaussian decay in the particle number makes each fixed derivative summable. The odd momentum constraint gives a positive gap for each individual particle integral, although that gap shrinks as the particle number increases.

Write $\mathcal{A}(v,w) = (v-w)/(1-vw)$ and $y_i = e^{\mathrm{i}q_i}$. Subtraction of two dispersion equations gives the canceled identity

$$\mathcal{A}(\rho_i,\rho_k)\mathcal{A}(y_i,y_k) = B(t)\left[\frac{\sin((q_i-q_k)/2)}{\sinh((\gamma_i+\gamma_k)/2)}\right]^2. \tag{5.13}$$

This is an identity of holomorphic canceled functions, including reciprocal y-pairs. The elementary identity

$$|1-vw|^2 - |v-w|^2 = (1-|v|^2)(1-|w|^2) + 4\Im v\,\Im w \tag{5.14}$$

and (5.4) imply $|\mathcal{A}(\rho_i,\rho_k)| \le \theta < 1$ at $t=0$. The same strict bound, with a slightly larger $\theta<1$, holds on a fixed complex parameter disk for all real angles. Every complete canceled pair in (5.13) is bounded there.

The Schur identity here is $\mathrm{Pf}(\mathcal{A}(y_i,y_k))_{i,k=1}^N = \prod_{i<k}\mathcal{A}(y_i,y_k)$ for even N. It follows by induction in N: as rational functions of y_N, the two sides have the same simple residues at $y_N = 1/y_k$. In checking the residues, the other paired factors cancel by $\mathcal{A}(a,b)\mathcal{A}(a,1/b)=1$, and the remaining product is the induction hypothesis. Their difference is constant because both are bounded at infinity, and that constant is zero at $y_N = y_{N-1}$. The identity then extends from generic tuples by rational continuation.

Apply this identity only to the y-variables, adjoining the ghost variable -1 because n is odd. The entry $\mathcal{A}(y_i,-1)$ is the removable constant one. Each of the $n!!$ matchings has $(n-1)/2$ real edges. Combine each matched y-factor with the corresponding ρ-factor by (5.13). The remaining $(n-1)^2/2$ ρ-factors are at most θ in modulus. Thus, with the one-body factors and normalization included,

$$\sup_{|t|<r_0,\ q\in\mathcal{T}_n} |H_n(t,q)| \le E_n, \qquad E_n = \exp\{-\kappa n^2 + Cn\log(n+1) + C\} \tag{5.15}$$

for some $\kappa>0$. Initially the Pfaffian calculation can be made off its apparent reciprocal-pair divisors; the canceled identity extends the estimate to those divisors. There is no division by a possibly zero ρ-factor. The same estimate holds when every angle is in a sufficiently small fixed complex neighborhood of π. The latter version will control the deformation of the local integration cycle.

Choose shortest lifts $q_i = \pi + \delta_i \bmod 2\pi$, with $\delta_i \in [-\pi,\pi]$, and put

$$S(q) = \sum_{i=1}^n \delta_i^2.$$

The odd momentum constraint says $\sum_i \delta_i = w\pi$, where w is odd. Therefore

$$S(q) \ge \frac{\pi^2}{n}. \tag{5.16}$$

On the inward ray, compactness of the branch and $\Im W(-\epsilon,\pi+\delta) \ge c(\epsilon + \mathrm{dist}(\delta,2\pi\mathbb{Z})^2)$ imply

$$\Re g(-\epsilon,\delta) \ge c\big(\epsilon + \mathrm{dist}(\delta,2\pi\mathbb{Z})^2\big). \tag{5.17}$$

Indeed $\Im W = \sinh(\Re g)\sin(\Im g)$, and the relevant roots remain in a fixed compact set. All parameter derivatives of g are bounded on a fixed complex neighborhood.

It follows from (5.16) that C_n is holomorphic on $|t| < c/n^2$. On this disk, $\Re\Gamma_n \ge c/n$ after decreasing c, and hence $|\coth(\Gamma_n/2)| \le Cn$. The torus volume is absorbed in the linear exponential in (5.15). Cauchy's estimate therefore gives

$$|[t^j]C_n(t)| \le (Cn^2)^j\, CnE_n. \tag{5.18}$$

More generally, the disk of radius $c(\epsilon + n^{-2})$ about $-\epsilon$ has $\Re\Gamma_n \geq c(n\epsilon + n^{-1})$. The same argument bounds the jth derivative at $-\epsilon$ by $j!(Cn^2)^j CnE_n$, uniformly for small ϵ. On compact subsets of $\mathcal{U}^-$, the kernel is bounded and (5.15) proves normal convergence. Dominated convergence for each fixed j now proves

$$a_j = \sum_{\substack{n\geq 1 \\ n \text{ odd}}} [t^j]C_n(t), \qquad \sum_n |[t^j]C_n(t)| < \infty. \tag{5.19}$$

5.3 An exact reduction to simple-pole periods

The shrinking gaps are encoded by nearby simple poles. We isolate their locations exactly and place the remaining analytic terms in a single remainder, whose coefficient growth is small enough for the later comparison.

Fix a sufficiently small $\sigma > 0$. There is a fixed $c_1 > 0$ such that on $|t| < c_1/n$ the part with $S \geq \sigma$ is analytic with a bounded kernel. To see this, use $\Re\Gamma_n(0,q) \geq cS$ and the perturbation bound $|\Gamma_n(t,q) - \Gamma_n(0,q)| \leq Cn|t|$. Choose c_1 to be a sufficiently small multiple of σ.

In the region $S < \sigma$, set

$$D_n(t,q) = \Gamma_n(t,q) - ni\phi.$$

The Taylor expansion of g at $(0,0)$ gives $|D_n(t,q)| \leq C(S + n|t|)$. If $p \nmid n$, the distance of $n\phi$ from $2\pi\mathbb{Z}$ is at least $2\pi/p$. Reduce σ, c_1 so that the perturbation is less than π/p. This part is also analytic, with bounded kernel, on $|t| < c_1/n$.

It remains to consider odd multiples of p and $S < \sigma$. Here $n\phi \in 2\pi\mathbb{Z}$, and

$$\coth(\Gamma_n/2) = \frac{2}{D_n} + \mathcal{R}(D_n),$$

where $\mathcal{R}$ is holomorphic on a fixed disk containing all these values for $|t| < c_1/n$. There is a unique small zero $t = T_n(q)$ of $D_n(t,q)$. This statement has a fixed-disk quantitative form. With $a_0 = B(0)/(\mathrm{i}s_\phi) \neq 0$, one has

$$\frac{1}{n}\partial_t D_n(t,q) = a_0 + O(|t| + \sigma)$$

on a fixed disk. By reducing that disk and σ, its distance from a_0 is less than $|a_0|/4$. Rouché's theorem on a circle of radius CS/n, followed by uniqueness from this derivative estimate, gives

$$|T_n(q)| \leq CS/n, \qquad D_n(t,q) = n(t - T_n(q))A_n(t,q), \tag{5.20}$$

where A_n is holomorphic on a fixed smaller disk and bounded away from zero. One can take

$$A_n(t,q) = \frac{1}{n}\int_0^1 \partial_t D_n(T_n(q) + u(t - T_n(q)), q)\ \mathrm{d}u.$$

All such disks and bounds are independent of n and q.

Put $J_n = H_n/A_n$. The exact identity

$$\frac{2H_n(t,q)}{D_n(t,q)} = \frac{2J_n(T_n(q),q)}{n(t - T_n(q))} + \frac{2}{n}\frac{J_n(t,q) - J_n(T_n(q),q)}{t - T_n(q)} \tag{5.21}$$

separates the pole from an analytic divided difference. The second term is bounded by CE_n, by fixed-disk Cauchy estimates. The $\mathcal{R}$ term has the same bound on $|t| < c_1/n$. Combining these

observations with the exterior and nonresonant regions gives

$$a_j = \sum_{\substack{n \text{ odd} \\ p|n}} A_{n,j}^{\mathrm{pole}} + R_j, \tag{5.22}$$

$$A_{n,j}^{\mathrm{pole}} = -\int_{S<\sigma} \frac{2H_n(T_n(q),q)}{\partial_t \Gamma_n(T_n(q),q)} T_n(q)^{-j-1}\, \mathrm{d}q, \tag{5.23}$$

$$|R_j| \le \sum_{n\ge 1} (Cn)^j \exp\{-\kappa n^2 + Cn\log(n+1) + C\} \tag{5.24}$$

$$\le \exp\{\tfrac{1}{2} j\log(j+2) + Cj + C\}. \tag{5.25}$$

For each n, the coefficient identity is first justified on the smaller disk $|t| < c/n^2$; no pole is crossed there. The last inequality follows by absorbing $Cn\log(n+1)$ into half the Gaussian and maximizing $j\log n - \kappa n^2/2$. This reduction isolates all small pole locations before any local change of angular variables is made.

5.4 Bounds for every winding and every particle index

Before evaluating the leading windings, we need estimates valid for every winding and particle index. These bounds will control the parts of the infinite sum that lie outside the asymptotic window.

The zero in (5.20) has a useful more precise estimate. Set

$$v_i = 1 - \cos\delta_i, \qquad \bar{v} = \frac{1}{n}\sum_i v_i.$$

At $t = \bar{v}$, the linear terms in the expansion of $\operatorname{arcosh}(b + B(\bar{v})(\bar{v} - v_i))$ cancel on averaging. The remaining average is bounded by $C\, n^{-1} \sum_i (v_i - \bar{v})^2$. The uniform nonzero derivative used in (5.20) therefore yields

$$T_n = \bar{v} + O\left(\frac{1}{n}\sum_i (v_i - \bar{v})^2\right).$$

Since $v_i \ge 0$, $\max_i v_i \le C\sigma$, and $\bar{v} = (S/(2n))(1 + O(\sigma))$, this implies

$$|T_n(q)| \ge (1 - C\sigma)\frac{S}{2n}, \qquad |\partial_t \Gamma_n(T_n(q),q)| \ge cn. \tag{5.26}$$

Retain the exact central numerator size

$$H_{0,n} = \frac{|B(0)|^{n^2/2}}{n!(2\pi)^{n-1} 2^{n(n-1)} s_\phi^{n^2}}. \tag{5.27}$$

Dividing the sine factors by their linear terms and all the hyperbolic sine factors by $\mathrm{i}s_\phi$ gives

$$|H_n(T_n(q),q)| \le H_{0,n}\Delta(\delta)^2 e^{CnS}, \qquad \Delta(\delta) = \prod_{i<k}(\delta_i - \delta_k). \tag{5.28}$$

Here is the relevant accounting. The logarithm of the anisotropy ratio costs $Cn^2|T_n| \le CnS$. The sum of the absolute one-body phase deviations is at most CS, so all pair denominators together cost CnS. Finally $\sum_{i<k} |\delta_i - \delta_k|^2 \le CnS$ controls the sine divided differences. Thus no undetermined constant to the power n^2 replaces the displayed normalization.

For an odd winding w, write

$$\delta_i = \frac{w\pi}{n} + \eta_i, \qquad \sum_i \eta_i = 0, \qquad Q(\eta) = \sum_i \eta_i^2, \qquad m_{n,w} = \frac{\pi^2 w^2}{n}. \tag{5.29}$$

Then $S = Q + m_{n,w}$ and $\Delta(\delta) = \Delta(\eta)$. The winding region is exactly $Q < \sigma - m_{n,w}$ when it is nonempty. In particular there are at most $C\sqrt{n}$ windings. Define

$$\alpha_n = \frac{n^2-1}{2}, \qquad P_n = \prod_{r=1}^{n-1} \frac{r!}{2^r}, \qquad G_n = \frac{n!\pi^{(n-1)/2} P_n}{\sqrt{n}}. \tag{5.30}$$

On the real zero-sum plane, with measure $\mathrm{d}\eta_1 \cdots \mathrm{d}\eta_{n-1}$,

$$\int e^{-Q(\eta)} \Delta(\eta)^2 \, \mathrm{d}\eta = G_n, \tag{5.31}$$

$$\int \frac{\Delta(\eta)^2}{(Q(\eta)+m)^{j+1}} \, \mathrm{d}\eta = G_n m^{\alpha_n - j - 1} \frac{\Gamma(j+1-\alpha_n)}{\Gamma(j+1)} \quad (j+1 > \alpha_n). \tag{5.32}$$

The unrestricted Gaussian Vandermonde integral is $n! \prod_{r=0}^{n-1} (\sqrt{\pi}\, r!/2^r) = n! \pi^{n/2} P_n$. This follows by writing the Vandermonde as a determinant of monic Hermite polynomials and integrating the determinant product. The monic squared norm is $\sqrt{\pi}\, r!/2^r$, obtained by integration by parts. In the decomposition $x_i = \eta_i + u$, $\sum \eta_i = 0$, the ordinary-coordinate Jacobian is n, and the mean integral is $\int e^{-nu^2} \, \mathrm{d}u = \sqrt{\pi/n}$. This proves (5.31). Homogeneity gives radial density proportional to $Q^{\alpha_n - 1} \, \mathrm{d}Q$, proving (5.32) by the ordinary beta integral.

Let $A^{\mathrm{pole}}_{n,j,w}$ denote the contribution of the wth winding to (5.23). Equations (5.26)–(5.32) imply

$$|A^{\mathrm{pole}}_{n,j,w}| \le \frac{C H_{0,n}}{n} e^{Cn\sigma} \left(\frac{2n}{1 - C\sigma} \right)^{j+1} G_n m_{n,w}^{\alpha_n - j - 1} \frac{\Gamma(j+1-\alpha_n)}{\Gamma(j+1)} \quad (j+1 > \alpha_n). \tag{5.33}$$

The dependence on w here is the explicit factor $|w|^{-2(j+1-\alpha_n)}$.

There is also a bound without any condition on j/α_n. Use $|T_n| \ge c/n^2$, and use homogeneity to write

$$\int_{Q<\sigma} \Delta(\eta)^2 \, \mathrm{d}\eta = G_n \frac{\sigma^{\alpha_n}}{\Gamma(\alpha_n + 1)}.$$

The elementary factorial bounds in (5.30) then give

$$\log \left(\sum_w |A^{\mathrm{pole}}_{n,j,w}| \right) \le 2j \log n - \frac{1}{2} n^2 \log n + Cn^2 + Cj + C. \tag{5.34}$$

In detail, $\log P_n = \frac{1}{2} n^2 \log n + O(n^2)$ and $\log \Gamma(\alpha_n + 1) = n^2 \log n + O(n^2)$; the remaining factors in $H_{0,n} G_n$ have logarithm $O(n^2)$. The number of windings costs only $O(\log(n+1))$. This proves (5.34) also when n is much larger than the Taylor order.

5.5 An exact symmetric pole coordinate

The next change of variables makes the pole location exactly quadratic while preserving the collision hyperplanes of the Vandermonde. This exactness is needed because the Taylor order grows with the particle number. The construction also keeps its domains and estimates independent of that dimension.

We now evaluate the first windings accurately enough for comparison between adjacent admissible particle numbers. All assertions in this subsection hold for sufficiently large odd multiples of p. Put

$$d_n = \frac{\pi}{n}, \qquad \tau_n = 1 - \cos d_n, \qquad B_n = B(\tau_n), \qquad V_n = \{v \in \mathbb{C}^n : \sum_i v_i = 0\}, \tag{5.35}$$

and use the $w = 1$ chart $\delta_i = d_n + v_i$. Distinguish the holomorphic quadratic form from the norm used in estimates:

$$Q(v) = \sum_i v_i^2, \qquad q(v) = \sum_i |v_i|^2.$$

All transposes and gradients in holomorphic identities on V_n use the complex bilinear extension of its real Euclidean structure. The integration measure remains $\mathrm{d}v_1 \cdots \mathrm{d}v_{n-1}$, with $v_n = -\sum_{i<n} v_i$.

Define

$$c_n = \frac{\mathrm{i}B_n}{2s_\phi}, \qquad h_n = \cos d_n - \frac{B_n b}{s_\phi^2} \sin^2 d_n = 1 + O(n^{-2}). \tag{5.36}$$

Direct differentiation, keeping the actual function $B(t)$, gives

$$g(\tau_n, d_n) = \mathrm{i}\phi, \qquad g_t(\tau_n, d_n) = -2c_n, \qquad \frac{1}{2} g_{\delta\delta}(\tau_n, d_n) = c_n h_n. \tag{5.37}$$

For example, $W_t = B(t) + B'(t)(t - 1 + \cos\delta)$; the B' term vanishes at this center. Also $g_{\delta\delta} = W_{\delta\delta}/\sinh g - (\cosh g) W_\delta^2 / \sinh^3 g$, which gives precisely (5.36).

The pole equation in this chart is

$$\sum_i \bigl(g(t, d_n + v_i) - \mathrm{i}\phi\bigr) = 0.$$

Its solution, denoted again by $T_n(v)$, is holomorphic and permutation invariant on a fixed ball $\|v\|_2 < r$, independent of n. Moreover

$$T_n(0) = \tau_n, \qquad |T_n(v) - \tau_n| \le \frac{Cq(v)}{n}, \qquad \|\nabla T_n(v)\|_2 \le \frac{C\sqrt{q(v)}}{n}, \qquad \|D^2 T_n(v)\| \le \frac{C}{n}. \tag{5.38}$$

Here and below operator norms are induced by the Euclidean norm. To prove these estimates, subtract the constant and linear terms of $g(t, d_n + v_i)$ in v_i. The linear terms sum to zero, and the remaining sum is bounded by $Cq(v)$. The t derivative of the sum is bounded below by cn, uniformly on this fixed neighborhood. Rouché's theorem, or the quantitative implicit-function theorem, gives the unique solution with $|T_n - \tau_n| \le Cq/n$. Uniqueness patches the local solutions over the whole ball. If $P = I - \mathbf{1}\mathbf{1}^T/n$, its gradient is

$$\nabla T_n = -\frac{P(g_\delta(T_n, d_n + v_i))_{i=1}^n}{\sum_i g_t(T_n, d_n + v_i)}.$$

The numerator has norm at most $C\|v\|_2$; another differentiation gives the Hessian bound. Both steps use scalar derivative bounds for g on fixed neighborhoods.

Introduce removable divided differences

$$\mathfrak{a}_n(t) = \frac{g(t, d_n) - \mathrm{i}\phi}{t - \tau_n}, \tag{5.39}$$

$$\mathfrak{e}_n(t, z) = g(t, d_n + z) - g(t, d_n) - g_\delta(t, d_n) z, \tag{5.40}$$

$$\mathfrak{h}_n(t, z) = -\frac{2\mathfrak{e}_n(t, z)}{h_n \mathfrak{a}_n(t) z^2}. \tag{5.41}$$

The function $\mathfrak{a}_n$ is uniformly bounded away from zero. Evenness of g in its second variable implies

$$\mathfrak{h}_n(\tau_n, 0) = 1, \tag{5.42}$$

$$|\mathfrak{h}_n(t,z) - 1| \le C\big(|t-\tau_n| + |d_n||z| + |z|^2\big), \tag{5.43}$$

$$|\partial_z \mathfrak{h}_n(t,z)| \le C\big(|d_n| + |z|\big), \qquad |\partial_z^2 \mathfrak{h}_n| + |\partial_t \mathfrak{h}_n| \le C. \tag{5.44}$$

Taylor's integral formula writes $\mathfrak{e}_n/z^2$ as an average of $g_{\delta\delta}(t, d_n + uz)$. Its first z derivative is an average of $g_{\delta\delta\delta}$, an odd function of its angular argument, and hence is $O(|d_n| + |z|)$. This proves the dimension-free bounds above.

Summing the divided differences at the pole gives the exact identity

$$T_n(v) - \tau_n = \frac{h_n}{2n} \sum_i v_i^2 \mathfrak{h}_n(T_n(v), v_i). \tag{5.45}$$

We turn its right-hand side into a quadratic form by an explicit permutation-equivariant change of variables. In particular, the construction never divides by $Q(v)$, which has a complex null cone.

First set

$$U_i(v) = v_i \sqrt{\mathfrak{h}_n(T_n(v), v_i)}, \qquad w = G(v) = PU(v), \tag{5.46}$$

using the square root near one. On a sufficiently small fixed ball,

$$\|DG - I\| \le C\big(|d_n|\|v\|_\infty + \|v\|_\infty^2 + q(v)/n\big) < \tfrac{1}{4}.$$

Thus G and its inverse are holomorphic on smaller fixed balls, with bounded derivatives. They commute with coordinate permutations. Because $T_n(v)$ is common to all components, one-variable divided differences give

$$\frac{U_i - U_k}{v_i - v_k} = 1 + \mathfrak{r}_{ik}(v), \tag{5.47}$$

$$|\mathfrak{r}_{ik}(v)| \le C\big(|T_n - \tau_n| + |d_n|(|v_i| + |v_k|) + |v_i|^2 + |v_k|^2\big). \tag{5.48}$$

The quotients extend holomorphically across $v_i = v_k$. Projection subtracts a common mean and does not change component differences. Summing logarithms of the near-one factors yields

$$\left|\log \frac{\Delta(G(v))}{\Delta(v)}\right| \le C\big(n^2|T_n - \tau_n| + n|d_n| \sum_i |v_i| + nq(v)\big) \le C\big(\sqrt{nq(v)} + nq(v)\big). \tag{5.49}$$

The quotient in this formula means its holomorphic extension.

The Jacobian estimate has no extra dimension loss. The derivative of G is a compressed diagonal matrix plus a rank-one matrix. The trace norm of the diagonal perturbation is bounded by

$$C\big(n|T_n - \tau_n| + |d_n| \sum_i |v_i| + q(v)\big),$$

and that of the rank-one perturbation is at most $Cq(v)/n$. The convergent matrix logarithm therefore gives

$$|\log \det_{V_n} DG(v)| \le C\big(|d_n|\sqrt{nq(v)} + q(v)\big). \tag{5.50}$$

The determinant on V_n equals the change-of-variable determinant in the constrained coordinates, since the same constant linear coordinate transformation is used on the source and target.

Restoring the discarded mean. Projection in (5.46) enforces the zero-sum constraint, but leaves a scalar mean in the quadratic energy. In the w coordinates let

$$\mu(w) = \frac{1}{n}\sum_i U_i(G^{-1}(w)).$$

At fixed t,

$$U_i = A_0(t)v_i + A_1(t)v_i^2 + A_2(t, v_i)v_i^3, \qquad A_1(t) = O(d_n),$$

with uniformly bounded analytic coefficients. The first terms sum to zero. Differentiating this expansion and using (5.38) gives, with $q = q(w)$,

$$|\mu| \le \frac{C}{n}(|d_n| + \sqrt{q})q, \qquad \|\nabla\mu\| \le \frac{C}{n}(|d_n| + \sqrt{q})\sqrt{q}, \qquad \|D^2\mu\| \le \frac{C}{n}(|d_n| + \sqrt{q}). \tag{5.51}$$

For the last assertion, the second derivative of G^{-1} is bounded by $C(|d_n| + \|w\|_2)$, as follows directly by differentiating (5.46). The chain-rule terms therefore obey the same bound.

Define the holomorphic vector field

$$\ell(w) = \int_0^1 \nabla\mu(uw)\, \mathrm{d}u \in V_n. \tag{5.52}$$

Then $\mu(w) = w^T\ell(w)$, and

$$\|\ell(w)\| \le \frac{C}{n}(|d_n| + \sqrt{q})\sqrt{q}, \qquad \|D\ell(w)\| \le \frac{C}{n}(|d_n| + \sqrt{q}).$$

Equivariance under the transposition of coordinates i, k shows that

$$\ell_i(w) - \ell_k(w) = (w_i - w_k)L_{ik}(w), \qquad |L_{ik}(w)| \le \frac{C}{n}(|d_n| + \sqrt{q}). \tag{5.53}$$

Indeed compare w with the vector with these two entries exchanged and integrate the derivative along the segment. The segment remains in the ball. The corresponding one-variable divided difference also proves holomorphic extension at the coincidence hyperplane.

Now set

$$R(w) = \left(I + n\ell(w)\ell(w)^T\right)^{1/2} w = w + \frac{n\,\mu(w)\ell(w)}{1 + \sqrt{1 + n\ell(w)^T\ell(w)}}. \tag{5.54}$$

The matrix square root is the branch near the identity. Its perturbation has norm at most $C(|d_n| + \sqrt{q})^2 q/n$. The map is uniformly biholomorphic on a smaller fixed ball. Exact multiplication gives

$$Q(R(w)) = Q(w) + n\mu(w)^2 = \sum_i U_i(v)^2. \tag{5.55}$$

This identity uses bilinear transposes and remains valid on the complex null cone.

By (5.53), the relative change of each pair difference under R is at most $C(|d_n| + \sqrt{q})^2 q/n$. The total logarithmic cost for the Vandermonde is therefore at most Cnq. Differentiating (5.54) shows that the trace norm of $DR - I$ is at most $C(|d_n| + \sqrt{q})^2 q$. For example, the term involving $D\ell$ has operator norm $O((|d_n| + \sqrt{q})^2 q/n)$; its trace norm costs at most the dimension $n - 1$. All the other terms are rank one. This proves the same bound for the logarithmic Jacobian.

The exact quadratic coordinate. Combining the projected map and the rank-one correction, define

$$\Psi_n(v) = \sqrt{h_n}\, R(G(v)). \tag{5.56}$$

It is a permutation-equivariant biholomorphism on fixed balls, $D\Psi_n(0) = \sqrt{h_n}I$, and it preserves all coincidence divisors. Equations (5.45) and (5.55) prove the exact normal form

$$T_n(v) = \tau_n + \frac{Q(\Psi_n(v))}{2n}. \tag{5.57}$$

After extracting the constant $h_n^{-\alpha_n}$, the Jacobian–Vandermonde ratio for its inverse has a holomorphic logarithm bounded by $C(\sqrt{nq} + nq)$. The exponent follows because the degree of Δ^2, plus the dimension of the plane, is $n(n-1) + (n-1) = 2\alpha_n$.

5.6 The complete residue density and its integration cycle

The normal form changes both the density and the integration cycle. We first control the complete residue density relative to its center value, then justify moving the transformed cycle back to a real plane without crossing a pole.

In the first-winding chart put

$$\varrho_n(v) = -\frac{2H_n(T_n(v), \pi + d_n + v)}{\partial_t \Gamma_n(T_n(v), \pi + d_n + v)}.$$

Here the arguments of H_n and Γ_n mean $q_i = \pi + d_n + v_i$, so that $\delta_i = d_n + v_i$. The quotient $\varrho_n/\Delta(v)^2$ is holomorphic and nonzero on a fixed small ball. Its value at zero is

$$\varrho_{n,0} = -\frac{2B_n^{\alpha_n - 1/2}}{n\, n!(2\pi)^{n-1} 2^{n(n-1)} s_\phi^{2\alpha_n}}. \tag{5.58}$$

At the center the numerator divided by Δ^2 has the factor $(\mathrm{i}s_\phi)^{-n^2}$. Since n is odd, $\mathrm{i}^{-n^2} = -\mathrm{i}$, while $\partial_t \Gamma_n = -\mathrm{i}nB_n/s_\phi$. The minus sign in (5.58) also includes the minus sign from the Taylor expansion of $(t - T_n)^{-1}$.

In the coordinates $w = \Psi_n(v)$, the complete residue form is

$$\varrho_n(v)\, \mathrm{d}v = \varrho_{n,0} h_n^{-\alpha_n} \Delta(w)^2 (1 + \mathcal{E}_n(w))\, \mathrm{d}w, \qquad |\mathcal{E}_n(w)| \le e^{C(\sqrt{nq(w)} + nq(w))} - 1. \tag{5.59}$$

Here the error is a relative error, after dividing by the explicit center value. To prove it, take logarithms only after canceling Δ^2. The anisotropy ratio costs $Cn^2|T_n - \tau_n|$. Moreover

$$\sum_i |g(T_n, d_n + v_i) - \mathrm{i}\phi| \le C\big(|d_n| \sum_i |v_i| + q(v) + n|T_n - \tau_n|\big).$$

Pair denominators cost at most Cn times this sum; the one-body denominators cost only the sum itself. The sine divided differences cost $C\sum_{i<k} |v_i - v_k|^2 \le Cnq(v)$. Finally the linear terms cancel in the derivative sum, giving

$$\frac{\partial_t \Gamma_n(T_n(v), \pi + d_n + v)}{n g_t(\tau_n, d_n)} = 1 + O(q(v)/n).$$

Together with (5.38) and the Jacobian–Vandermonde estimates, these are exactly the logarithmic bound in (5.59).

We next justify replacing the original real cycle by a real cycle in the exact w coordinates. Choose a small fixed radius $a > 0$. For large n, the real ball $Q(v) < a^2$ lies strictly in the

first-winding region $Q(v) < \sigma - \pi^2/n$. The omitted shell has $|T_n| \geq c/n$, by (5.26), and hence its contribution to the jth coefficient is bounded by

$$(Cn)^j \exp\{-\kappa n^2 + Cn\log(n+1) + C\}.$$

Use (5.15) for the numerator and absorb the volume and other factors into the indicated constants.

By choosing a small and then n large, one has $\|\Psi_n(v) - v\|_2 \leq \eta\|v\|_2$, with a fixed $\eta < 1/4$, on a slightly larger ball. For real v, the straight homotopy $w_u = (1-u)v + u\Psi_n(v)$ satisfies

$$\Re Q(w_u) \geq (1 - 2\eta - \eta^2)Q(v) > 0 \quad (v \neq 0). \tag{5.60}$$

This follows by writing $w_u = v + e$, $\|e\| \leq \eta\|v\|$, and estimating $2v^T e + e^T e$. The pole $\tau_n + Q(w)/(2n) = 0$ is therefore avoided throughout the homotopy.

Apply Stokes' theorem to the holomorphic $(n-1)$-form

$$\varrho_{n,0} h_n^{-\alpha_n} \frac{\Delta(w)^2(1 + \mathcal{E}_n(w))}{(\tau_n + Q(w)/(2n))^{j+1}}\, \mathrm{d}w_1 \wedge \cdots \wedge \mathrm{d}w_{n-1}.$$

The difference between its integrals over the image of the real ball and over the real ball itself is the integral over the lateral boundary of the homotopy. On that boundary, (5.60) gives $|\tau_n + Q(w)/(2n)| \geq c/n$. The inverse image under Ψ_n remains in a fixed complex ball around the all-endpoint configuration. The complex-angle version of (5.15) bounds its complete numerator. The implicit derivative is at least cn; the Jacobians and the parameterized boundary volume cost at most $\exp(Cn\log(n+1) + C)$. For the latter bound it is enough to use bounded operator norms for the maps and their inverses and the elementary Euclidean ball and sphere volume bounds in dimension $n-1$. Consequently the entire boundary error has the bound

$$|b_{n,j}| \leq (Cn)^j \exp\{-\kappa n^2 + Cn\log(n+1) + C\}. \tag{5.61}$$

5.7 A uniform beta evaluation of the first two windings

The real cycle permits an exact beta integral. The task is to show that the relative density error remains small in the simultaneous large-particle, high-Taylor-order limit. The two minimal windings then have the same complex coefficient and add.

For $j + 1 > \alpha_n$, (5.32) yields

$$\int_{V_n \cap \mathbb{R}^n} \frac{\Delta(w)^2}{(\tau_n + Q(w)/(2n))^{j+1}}\, \mathrm{d}w = G_n (2n)^{\alpha_n} \tau_n^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j+1)}. \tag{5.62}$$

The exact gamma quotient controls relative errors in the regime below, where replacing it by $j^{-\alpha_n}$ incurs a logarithmic error of order α_n^2/j.

We prove that (5.59) has relative integrated error

$$O\left(\frac{n}{\sqrt{j}} + \frac{n^2}{j}\right) \quad \text{as } n \longrightarrow \infty, \quad n^2/j \longrightarrow 0. \tag{5.63}$$

Under the normalized positive measure in (5.62), the variable $x = Q/(2n\tau_n)$ has density proportional to

$$x^{\alpha_n - 1}(1+x)^{-j-1}\, \mathrm{d}x.$$

Since $nQ = 2n^2\tau_n x \asymp x$, the error in (5.59) is bounded by $e^{C(\sqrt{x}+x)} - 1$. On $0 \le x \le \delta$, for any fixed small $\delta > 0$, this is at most $C(\sqrt{x} + x)$. The exact mean

$$\mathbb{E}x = \frac{\alpha_n}{j - \alpha_n}$$

and Cauchy–Schwarz show that its integral is bounded by $C(\sqrt{\alpha_n/j} + \alpha_n/j)$.

It remains to estimate the tail, where the preceding exponential cannot be integrated over the whole half-line. The fixed local ball has $x \le Ln$ for a constant L. Uniformly on $\delta \le x \le Ln$, when $n^2/j \to 0$,

$$C(\sqrt{x} + x) \le \frac{j}{4} \log(1 + x).$$

Write $Z(\alpha, j) = \Gamma(\alpha)\Gamma(j + 1 - \alpha)/\Gamma(j + 1)$. The tail integral divided by the original normalization is at most

$$(1 + \delta)^{-j/4} \frac{Z(\alpha_n, j/2)}{Z(\alpha_n, j)}.$$

Because α_n is an integer,

$$\frac{Z(\alpha_n, j/2)}{Z(\alpha_n, j)} = \prod_{r=0}^{\alpha_n - 1} \frac{j - r}{j/2 - r} \le 4^{\alpha_n} \qquad (\alpha_n \le j/4).$$

Thus this tail is $\exp(-cj + O(\alpha_n))$. The same bound, without the exponential error factor, controls extending the main integral from the fixed real ball to the whole real plane. This proves (5.63) with constants uniform in the specified regime.

The winding $w = -1$ is identical to $w = 1$: replace all lifts by their negatives, use evenness of g, and use the squared sine factors. The ordinary real measure has the same orientation since $n - 1$ is even. The two leading terms therefore add. Substituting (5.58) and (5.30) into (5.62) gives the following exact-factor model:

$$A^{\mathrm{pole}}_{n,j,1} + A^{\mathrm{pole}}_{n,j,-1} = M(n, j)(1 + e_{n,j}) + b_{n,j}, \tag{5.64}$$

$$|e_{n,j}| \le C \left(\frac{n}{\sqrt{j}} + \frac{n^2}{j} \right), \tag{5.65}$$

$$M(n, j) = -\frac{2^{2-\alpha_n} B_n^{\alpha_n - 1/2} P_n n^{\alpha_n - 3/2}}{\pi^{(n-1)/2} s_\phi^{2\alpha_n}} \, h_n^{-\alpha_n} \tau_n^{\alpha_n - j - 1} \frac{\Gamma(j + 1 - \alpha_n)}{\Gamma(j + 1)}. \tag{5.66}$$

The term $b_{n,j}$ satisfies (5.61). The power $B_n^{\alpha_n - 1/2}$, including its varying complex phase, is part of the model.

5.8 Exact finite differences and a single dominant index

We compare the moduli of the exact-factor models at neighboring admissible particle numbers. Strict discrete concavity isolates a single maximum, and the Taylor order is chosen so that the selected index sits at that maximum. This yields noncancellation without any assumption on the phases of the other indices.

Let

$$L(n, j) = \log |M(n, j)|, \qquad d = 2p.$$

The admissible indices are the odd multiples of p, so their spacing is d. Elementary Stirling bounds in (5.66) give, as $n \to \infty$ and $n^2/j \to 0$,

$$L(n,j) = 2j\log n - \frac{n^2}{2}\log j + c_* n^2 - j\log(\pi^2/2) + O(n\log n + n^4/j + j/n^2). \tag{5.67}$$

For example,

$$\log P_n = \frac{1}{2}n^2\log n - \left(\frac{3}{4} + \frac{1}{2}\log 2\right)n^2 + O(n\log n), \qquad \log\tau_n = \log(\pi^2/2) - 2\log n + O(n^{-2}),$$

while the logarithm of the gamma quotient is $-\alpha_n \log j + O(\alpha_n^2/j)$. These formulas give exactly the value of c_* in (5.12).

The adjacent-index comparison requires finite differences of L. We compute them from the exact factors, since the error bound in (5.67) gives no derivative control. The resulting identity is

$$L(n+d,j) - 2L(n,j) + L(n-d,j) = d^2\left(-\frac{2j}{n^2} - \log j + 2c_*\right) + O_d(n^2/j + j/n^4 + n^{-2}). \tag{5.68}$$

Here is a direct verification with the required uniform remainder. First,

$$\log P_{n+d} - 2\log P_n + \log P_{n-d} = \sum_{k=0}^{d-1}\sum_{\ell=1}^{d}\log(n-d+k+\ell) - d^2\log 2 \tag{5.69}$$

$$= d^2(\log n - \log 2) + O_d(n^{-2}). \tag{5.70}$$

The offsets in this double sum have sum zero. Next, because α_n is an integer, for real $j > \alpha_{n+d}$ the exact identity is

$$\log\frac{\Gamma(j+1-\alpha_n)}{\Gamma(j+1)} = -\alpha_n\log j + \sum_{r=0}^{\alpha_n-1} -\log(1-r/j).$$

The upper and lower adjacent increments of α_n have lengths $dn + d^2/2$ and $dn - d^2/2$. Pair the $dn - d^2/2$ corresponding summands; their arguments differ by $O_d(n)$, and the derivative of $-\log(1-x/j)$ is at most $1/(j-\alpha_{n+d})$. Their total difference is $O_d(n^2/j)$. The d^2 unpaired summands cost the same amount. Thus this factor contributes $-d^2\log j + O_d(n^2/j)$.

For a precise accounting of all remaining factors, put

$$c_\tau = \pi^2/2, \qquad r_n = \log\tau_n - \log c_\tau + 2\log n, \qquad K = -\tfrac{1}{2}\log 2 + \tfrac{1}{2}\log|B(0)| - \log s_\phi + \tfrac{1}{2}\log c_\tau.$$

Up to a constant independent of n, j, the logarithm $L(n,j)$ is exactly

$$\begin{aligned} &2j\log n - j\log c_\tau - \alpha_n\log j + \log P_n - \tfrac{1}{2}n^2\log n + Kn^2 - \tfrac{n}{2}\log\pi + \log n \\ &\quad + \sum_{r=0}^{\alpha_n-1} -\log(1-r/j) + (\alpha_n - \tfrac{1}{2})\Re\log(B_n/B(0)) - \alpha_n\Re\log h_n + (\alpha_n - j - 1)r_n. \end{aligned}$$

The last three terms have convergent expansions in n^{-2}, and

$$r_n = -\frac{\pi^2}{12n^2} + O(n^{-4}).$$

Their second differences are $O_d(n^{-2} + j/n^4)$, as is seen either term by term in these convergent expansions or by differentiating them on a fixed neighborhood of $n^{-1} = 0$. Also

$$\Delta_d^2(-\tfrac{1}{2}n^2\log n) = -d^2(\log n + \tfrac{3}{2}) + O_d(n^{-2}), \qquad \Delta_d^2(2j\log n) = -\frac{2d^2 j}{n^2} + O_d(j/n^4).$$

Combine these with (5.70) and $2c_* = 2K - \frac{3}{2} - \log 2$. This proves (5.68) directly from the exact model.

In particular, throughout the intermediate range

$$\log j < n < \frac{\sqrt{j}}{(\log j)^{1/4}}, \tag{5.71}$$

the second difference is at most $-cd^2 \log j$ for all sufficiently large j, whenever the neighboring indices are also in that range. Indeed $n^2/j \to 0$ uniformly, and j/n^4 is smaller than j/n^2 by a factor at most $(\log j)^{-2}$. This is concavity over the entire intermediate range.

Fix a large admissible n, and temporarily let j be real of order $n^2 \log n$. Set

$$f_n(j) = L(n+d, j) - L(n-d, j).$$

The two gamma arguments differ by the integer $2dn$. Differentiating their exact product gives

$$f_n'(j) = \log \frac{\tau_{n-d}}{\tau_{n+d}} - \sum_{r=1}^{2dn} \frac{1}{j + 1 - \alpha_{n-d} - r} \tag{5.72}$$

$$= \frac{4d}{n} - \frac{2dn}{j} + O_d(n^{-3} + n^3/j^2) > 0. \tag{5.73}$$

Exact first differences also give

$$\frac{f_n(j)}{2d} = \frac{2j}{n} - n \log j + 2c_* n + O_d(\log n + n^3/j + j/n^3). \tag{5.74}$$

For detail, the quotient P_{n+d}/P_{n-d} is a product of $2d$ factorials; Stirling's bound on these individual factorials gives its centered first difference $n \log n - (1 + \log 2)n + O_d(\log n)$. The centered first difference of the gamma correction sum is $O_d(n^3/j)$. The convergent n^{-2} expansions above contribute $O_d(n^{-3} + j/n^3)$. The displayed leading terms then follow exactly as in the second-difference calculation.

At

$$j = n^2 \left(\log n + \tfrac{1}{2} \log \log n - c_* \pm \epsilon \right)$$

the expression in (5.74) has opposite signs for each fixed $\epsilon > 0$ and all large n. There is a unique crossing in this interval by (5.73). Round it to the nearest integer j_n. We obtain

$$j_n = n^2 \left(\log n + \tfrac{1}{2} \log \log n - c_* + o(1) \right), \qquad f_n(j_n) = O_d(1/n). \tag{5.75}$$

These n lie strictly inside (5.71).

Let $\lambda_k = L(n + kd, j_n)$ for the admissible indices in that range. Their successive first differences decrease by at least $cd^2 \log j_n$. Equation (5.75) gives $\lambda_1 - \lambda_{-1} = O_d(1/n)$. Consequently

$$\lambda_0 - \lambda_{\pm 1} \geq c' d^2 \log j_n, \qquad \lambda_k - \lambda_0 \leq -c'' d^2 k^2 \log j_n + O_d(|k|/n) \quad (k \neq 0).$$

Summing this discrete Gaussian proves

$$\sum_{\substack{m \neq n,\ m \text{ odd},\ p \mid m \\ m \text{ satisfies (5.71)}}} |M(m, j_n)| \leq |M(n, j_n)|\, O(j_n^{-c''' d^2}). \tag{5.76}$$

The complex phases of the different models play no role in this inequality.

5.9 Exhausting the remaining indices and windings

The comparison must cover the complete coefficient, including indices far from the selected one. We divide the remainder into small indices, analytic and cycle errors, the infinite large-index tail, higher windings, and the other indices in the intermediate range. Each class is compared with the same selected model.

We compare every part of (5.19) with the selected model. First, (5.67) and (5.75) imply

$$\log|M(n,j_n)| = j_n\log j_n - j_n\log\log j_n + j_n\big(\log(4/\pi^2)-1\big) + o(j_n), \tag{5.77}$$

$$|M(n,j_n)|^{1/j_n} \sim \frac{4j_n}{e\pi^2\log j_n}. \tag{5.78}$$

Indeed (5.75) gives

$$n^2 = \frac{j_n}{\frac{1}{2}\log j_n - c_* + o(1)}.$$

Substituting this in the four leading terms of (5.67) gives (5.77); each error there is $o(j_n)$.

The following estimates cover all the complementary terms.

Small particle indices. For $m \le \log j$, the full coefficient bound (5.18) gives

$$\sum_{\substack{m\le\log j\\ m \text{ odd}}} |[t^j]C_m(t)| \le \exp\{2j\log\log(j+2) + Cj + C\}. \tag{5.79}$$

This is negligible compared with (5.77) when $j = j_n$. Equivalently, after the pole reduction one can apply the same bound to the poles plus their analytic remainders, since their difference has the smaller bound (5.25).

All analytic and cycle remainders. The sum of the regular remainders in (5.25) and all the terms (5.61) is bounded by

$$\exp\{\tfrac{1}{2}j\log(j+2) + Cj + C\}. \tag{5.80}$$

This too is negligible at the selected orders. This estimate includes the entire nonresonant particle class $p \nmid m$ outside the small-index range.

All sufficiently large particle indices. For

$$m \ge \frac{\sqrt{j}}{(\log j)^{1/4}},$$

use (5.34), which has no restriction on m^2/j. With $x = m^2$, its exponent is at most

$$j\log x - \frac{1}{4}x\log x + Cx + Cj + C.$$

Its derivative in x is negative on $x \ge j/\sqrt{\log j}$, for all large j, and at the lower endpoint its value is

$$j\log j - \frac{1}{2}j\log\log j - \frac{1}{4}j\sqrt{\log j} + o(j\sqrt{\log j}). \tag{5.81}$$

For example its derivative is at most $-c\log j$ there for large j, since $j/x \le \sqrt{\log j}$ and $\frac{1}{4}\log x \sim \frac{1}{4}\log j$. Summing the resulting decreasing exponential over the discrete values $x = m^2$ costs at most a harmless polynomial factor. Thus (5.81) bounds the logarithm of the whole infinite large-index tail up to $o(j\sqrt{\log j})$. Its gap below (5.77) tends to infinity faster than j.

Higher odd windings in the intermediate range. Let $M_0(m,j) > 0$ be the modulus of the expression (5.66) with B_m replaced by $B(0)$, h_m by 1, and τ_m by $\pi^2/(2m^2)$. The exact factors give

$$\left|\log \frac{M_0(m,j)}{|M(m,j)|}\right| \le C(1 + j/m^2). \tag{5.82}$$

For the τ factor this follows from $\log(\tau_m/(\pi^2/(2m^2))) = O(m^{-2})$; the B and h factors cost $O(1)$. Comparison of (5.33) with this explicit M_0 yields, for each odd $|w| \ge 3$,

$$|A^{\mathrm{pole}}_{m,j,w}| \le C M_0(m,j) e^{Cm\sigma}(1 - C\sigma)^{-j-1}|w|^{-2(j+1-\alpha_m)}.$$

The constants C here do not depend on m, j, w. Choose the fixed σ still smaller, if necessary, so that $-\log(1 - C\sigma) < \frac{1}{2}\log 3$. Throughout (5.71),

$$\alpha_m = o(j), \qquad m = o(j), \qquad j/m^2 = o(j)$$

uniformly. The number of possible windings is at most $C\sqrt{m}$. Hence for some $c > 0$,

$$\sum_{\substack{w \text{ odd} \\ |w|\ge 3}} |A^{\mathrm{pole}}_{m,j,w}| \le |M(m,j)| e^{-cj} \tag{5.83}$$

uniformly in that range. The choice of a smaller σ merely changes the fixed constants in the earlier arguments.

The first windings of every intermediate index. Equation (5.65) tends uniformly to zero in (5.71); it is at most $C(\log j)^{-1/4}$. At $j = j_n$, (5.76) shows that the sum of the moduli of all intermediate models other than $M(n, j_n)$ is $o(|M(n, j_n)|)$. The sum of all their relative errors is bounded by

$$o(1) \sum_m |M(m, j_n)| = o(|M(n, j_n)|).$$

The same is true of all higher windings by (5.83).

Substitution of the estimates for all odd indices, windings and remainders into (5.19) gives

$$a_{j_n} = M(n, j_n)(1 + o(1)). \tag{5.84}$$

5.10 Nonextension and uniformity

Equations (5.78) and (5.84) prove (5.11), so $\limsup_j |a_j|^{1/j} = \infty$. If F admitted a holomorphic extension to a disk about zero from the physical side $\mathcal{U}^-$, its derivatives on the inward negative ray would agree with those of that extension. By (5.9), its Taylor coefficients at zero would therefore be the a_j. Cauchy's estimate on any smaller extension disk would bound $|a_j| \le CR^{-j}$, a contradiction. If $P(0) \ne 0$, a holomorphic extension of PF would, after division by the nonvanishing holomorphic factor P, give an extension of F. Thus a local physical normalization by such a factor cannot remove the obstruction.

Finally, all the constants used above arise from a common holomorphic neighborhood for B, a lower bound for $-\Im B(0)$, and fixed scalar bounds for the dispersion and its derivatives. A holomorphic family with $\Im B(0, \eta) \le -\beta < 0$ on a relatively compact parameter polydisk supplies these uniformly by Cauchy's estimate. The square roots and logarithms near one used in the normal form also have common neighborhoods. The exact model uses $B(\tau_n, \eta)$. The crossing and range estimates are uniform on each such compact family, with $c_* = c_*(\eta)$ and selected orders $j_n(\eta)$. This completes the proof of Theorem 5.1.

6 The isotropic restriction: a unified even–odd proof

On the collinear isotropic line we use the full form-factor integrals directly. The analytic estimates treat both phases with their corresponding particle parities.

Write ζ for the auxiliary exterior variable, and put

$$E(\zeta) = \zeta + \zeta^{-1}, \qquad D(x, y; \zeta) = E(\zeta) - \tfrac{1}{2}(x + x^{-1} + y + y^{-1}).$$

For the low-temperature restriction $\zeta = s$; for the high-temperature restriction $\zeta = s^{-1}$. Define

$$T_N^\sigma(\zeta) = \frac{1}{N!} \oint \cdots \oint \frac{Z^{-1} + Y^{-1}}{(1 - \sigma Z)(1 - \sigma Y)} \prod_{i<j} P_{ij} \prod_i R(y_i; \zeta) \frac{\mathrm{d}y_i}{2\pi \mathrm{i}}, \tag{6.1}$$

$$f_N(\zeta) = \frac{1}{N!} \oint \cdots \oint (ZY)^{-1} \prod_{i<j} P_{ij} \prod_i R(y_i; \zeta) \frac{\mathrm{d}y_i}{2\pi \mathrm{i}}, \tag{6.2}$$

where $\sigma \in \{1, -1\}$, $Y = \prod_i y_i$, $Z = \prod_i z_i$, and

$$W = E(\zeta) - \tfrac{1}{2}(y + y^{-1}), \quad z = W - \sqrt{W^2 - 1}, \quad R = \frac{2z^2}{1 - z^2}, \quad P_{ij} = -\frac{(y_i - y_j)^2 z_i z_j}{y_i y_j (1 - z_i z_j)^2}.$$

On real $\zeta > 1$ the root satisfies $0 < z < 1$ on the angular circle; elsewhere it is the continued interior root. Put $T_N = T_N^1$ and $C_N = f_N + 2T_N$.

These expressions retain the full-site normalization. Their double-contour lift, with $X = \prod_i x_i$, is

$$T_N^\sigma = \frac{1}{N!} \oint \cdots \oint \frac{X^{-1} + Y^{-1}}{(1 - \sigma X)(1 - \sigma Y)} \prod_{i<j} \frac{x_i - x_j}{1 - x_i x_j} \frac{y_i - y_j}{1 - y_i y_j} \prod_i \frac{\mathrm{d}x_i \, \mathrm{d}y_i}{(2\pi \mathrm{i})^2 D(x_i, y_i; \zeta)}. \tag{6.3}$$

For f_N replace the first rational factor by $(XY)^{-1}$. Choose a common radius $r < 1$ with $|z(y; \zeta)| < r$ on the y circle. The only enclosed dispersion pole is then $x = z$, with residue $2z^2/(1 - z^2)$. At $x = 0$, $D^{-1} = -2x + O(x^2)$ cancels the apparent numerator pole. All pair and product poles remain outside after each successive residue. Subtracting two dispersion equations gives

$$(z_i - z_j)(1 - z_i z_j) = -\frac{z_i z_j}{y_i y_j}(y_i - y_j)(1 - y_i y_j),$$

and hence the canceled expression for P_{ij}. The full lattice multiplier is exactly

$$\frac{(1 + X)(1 + Y)}{XY(1 - X)(1 - Y)} = \frac{1}{XY} + \frac{2(X^{-1} + Y^{-1})}{(1 - X)(1 - Y)}. \tag{6.4}$$

Thus f_N is the onsite coefficient, T_N is the indicated half of the offsite coefficient, and C_N is the full lattice coefficient. Equivalently,

$$C_N(\zeta) = \frac{1}{N!} \int_{\sum q_i = 0 \ (2\pi)} \prod_i \frac{1}{\sinh \gamma_i} \frac{1 + Z}{1 - Z} \prod_{i<j} \left[\frac{\sin((q_i - q_j)/2)}{\sinh((\gamma_i + \gamma_j)/2)} \right]^2 \prod_{i<N} \frac{\mathrm{d}q_i}{2\pi}, \tag{6.5}$$

where $\cosh \gamma_i = E(\zeta) - \cos q_i$ and $z_i = e^{-\gamma_i}$. The independent measure has $N - 1$ coordinates, with $q_N = -\sum_{i<N} q_i$ modulo 2π. On real $\zeta > 1$ let $q_N(m, n)$ be $1/N!$ times the N-angle integral with multiplier $Z^m Y^n$, measure $\prod_i \mathrm{d}q_i/(2\pi \sinh \gamma_i)$, and the displayed squared pairs. Angle reversal gives $q_N(m, n) = q_N(m, -n)$; exchanging the two coordinate sets in (6.3) gives $q_N(m, n) = q_N(n, m)$ for $m, n \geq 0$. Any permutation sign of the top form is canceled by the same permutation of the product

orientation, for either parity. A common small angle strip has $\Re\gamma_i \geq c > 0$, so simultaneous angle shifts give, at fixed N, ζ, a bound $C_{N,\zeta} e^{-N(cm+\eta|n|)}$. Consequently the spatial sum is absolutely convergent. Summing $n \in \mathbb{Z}$ enforces $\sum q_i = 0$ modulo 2π, and summing the reflected m coordinate gives $(1+Z)/(1-Z)$. Alternatively,

$$\frac{Y+Z}{(1-Y)(1-Z)} = \sum_{m\geq 1, n\geq 0} Y^m Z^n + \sum_{m\geq 0, n\geq 1} Y^m Z^n$$

counts each positive axis once and each open positive quadrant twice. Thus $f_N + 2T_N$ counts exactly every lattice site, establishing (6.5) without omitting the onsite term.

6.1 Compact convergence and a dense arithmetic family

Compact convergence defines the holomorphic sum inside its domain. The arithmetic construction then selects boundary points where one particle order is the first possible resonance and its two collision charts are uniquely determined.

For even N the Schur identity states that

$$\operatorname{Pf}\left[\frac{v_i - v_j}{1 - v_i v_j}\right] = \prod_{i<j} \frac{v_i - v_j}{1 - v_i v_j}.$$

For odd N, append $v_{N+1} = -1$; its last upper column is the removable constant one, and the same identity gives the product over the N real indices. One proof puts $a_i = (1+v_i)/(1-v_i)$ and compares residues at $a_N = -a_i$ inductively. Both rational functions have the same residues, are bounded at infinity, and vanish at $a_N = a_{N-1}$.

On a compact exterior neighborhood choose fixed radii $r < 1$ with all interior roots bounded by $q < r$ and bounded away from zero. The global inverse products cost only C^N. Hadamard's inequality gives $|\operatorname{Pf} A| \leq C^{m/2}(m-1)^{m/4}$ for an even matrix of size m. Applying this to both matrices, including the ghost index if necessary, gives

$$|T_N^\sigma| + |f_N| + |C_N| \leq C_0 \frac{C_1^N N^{(N+1)/2}}{N!}. \tag{6.6}$$

The bound is summable. Cauchy's estimate on a larger compact neighborhood proves normal convergence of every fixed derivative on compact exterior sets.

Lemma 6.1 (The selected first orders)**.** *Let $p \geq 11$ be prime, choose distinct $a, b \in \{1, \ldots, (p-1)/2\}$, and set*

$$\alpha = 2\pi a/p, \quad \beta = 2\pi b/p, \quad E_* = \cos\alpha + \cos\beta = 2\cos\theta, \quad \zeta_* = e^{\mathrm{i}\theta}.$$

For the even family choose $\alpha, \beta \in (0, \pi/2)$ and put $N_0 = 2p$, $k = N_0^2/2 - 1$. For the odd family choose both angles in $(0, \pi/2)$ or both in $(\pi/2, \pi)$ and put $N_0 = p$, $k = (p^2-1)/2$. The first order of the respective parity satisfying a Nickel equation is N_0, and its unordered cosine pair is uniquely $\{\cos\alpha, \cos\beta\}$. No endpoint configuration is resonant at any order. The even family is dense in the upper-right arc; the odd families are dense in both upper quadrants.

Proof. Let $\omega = e^{2\pi\mathrm{i}/p}$, $d = (p-1)/2$, $u_j = \omega^j + \omega^{-j}$, and $F = \mathbb{Q}(\omega + \omega^{-1})$. Every nonzero-index u_j has trace -1. Thus $v = u_a + u_b$ has trace -2 and is irrational: if rational it would be an integer, whereas its trace would force $v = -2/d$. If $p \nmid n$, then $\mathbb{Q}(\omega) \cap \mathbb{Q}(e^{2\pi\mathrm{i}/n}) = \mathbb{Q}$. Indeed the compositum has degree $\varphi(pn) = \varphi(p)\varphi(n)$, so the intersection has degree one. A lower order of the required parity would contradict this fact.

The only rational relation between $1, u_1, \dots, u_d$ is a multiple of $1 + \sum_j u_j = 0$: lift the relation to a polynomial of degree at most $p-1$ and use the minimal polynomial $1 + X + \cdots + X^{p-1}$. At order p all cosines are $u_j/2$ or 1; at order $2p$ they are $\pm u_j/2$ or ± 1. An equality of two cosine pairs has at most four nonconstant positions. Since $d \geq 5$, a zero coefficient forces the multiple of the relation to vanish, proving uniqueness.

For $E_* > 0$, an endpoint resonance requires a root of unity ξ with $\xi + \xi^{-1} = v - 2$. Its trace is $-2 - 2d < -2d$, impossible because every conjugate of a root-of-unity trace belongs to $[-2, 2]$. For $E_* < 0$ the possible endpoint equation is $\xi + \xi^{-1} = v + 2$. Direct expansion gives $\mathrm{Tr}_F(v^2) = 4d - 6$. If all conjugates of $v + 2$ were in $[-2, 2]$, every conjugate of v would be in $[-4, 0]$, where $v^2 \leq -4v$. Taking traces would imply $4d - 6 \leq 8$, contrary to $d \geq 5$. Approximating any angle in either relevant half interval by two distinct adjacent prime-grid angles proves density. □

Lemma 6.2 (Exponential endpoint separation)**.** *If a fixed algebraic unit-modulus number ξ is not a root of unity, there is $A > 0$ such that $|1 - \xi^N|, |1 + \xi^N| \geq e^{-AN}$ for every positive integer N.*

Proof. Write its primitive irreducible polynomial as $a_d \prod_{\nu=1}^{d} (X - \xi_\nu)$ with $\xi_1 = \xi$. Its nonzero integer resultant with $X^N - 1$ has modulus at least one. Bounding every other conjugate by $|\xi_\nu^N - 1| \leq 2 \max(1, |\xi_\nu|)^N$ proves the first assertion. The inequality $|1 + \xi^N| \geq |1 - \xi^{2N}|/2$ proves the second after increasing A. □

6.2 The first complete singular coefficient

At a selected point, the local calculation has two parts: bounded contributions from the complementary configurations, and explicit residues from the two collision charts. Their complex phases must be matched before the leading coefficient can be used for the full sum.

Put $\zeta_\varepsilon = (1 + \varepsilon)\zeta_*$ and keep the integration radius $r = e^{-c_0 \varepsilon}$ fixed during every complex derivative. On a sufficiently small disk of radius $c\varepsilon$, $\Im W \geq c'\varepsilon$ for every angle. The interior roots belong to the lower unit half-disk and satisfy $|z| < r$ if c_0 is small enough. These statements hold for either upper quadrant.

Lemma 6.3 (Fixed-order complements)**.** *At each point in Lemma 6.1, every fixed derivative of each lower coefficient of the selected parity is bounded on the exterior radial approach. At order N_0, the complement of the two lower double-contour configurations*

$$(x_i, y_i) = (e^{-\mathrm{i}\beta}, e^{-\mathrm{i}\alpha}) \quad \textit{for all } i, \qquad \textit{and their exchange}$$

has bounded derivatives of every fixed order. The onsite coefficient has bounded fixed derivatives for every fixed N.

Proof. The potentially singular factors on the unit double torus have gradient vectors consisting of the two positive all-ones product vectors, the positive reciprocal-pair vectors, and the dispersion vectors with entries $(\Im x_i, \Im y_i)$. The sign of dispersion damping is fixed by $\Im E(\zeta_\varepsilon) > 0$. A nonnegative zero combination cannot contain a pair vector: its two positive entries would require both corresponding imaginary parts to be negative, whereas reciprocal circle coordinates have opposite imaginary parts.

If both product vectors occur, every coordinate is lower and the ratios $\sin A / \sin B$ agree along $\cos A + \cos B = E_*$. Along that curve

$$\frac{\mathrm{d}}{\mathrm{d}A} \frac{\sin A}{\sin B} = \frac{E_*(1 - \cos A \cos B)}{\sin^3 B}$$

has a strict constant sign. All pairs therefore coincide, and both common coordinates are Nth roots of unity. Lemma 6.1 leaves exactly the specified configurations at N_0 and none below it. With only one product vector, the other coordinates must all be $+1$ when $E_* > 0$, or all -1 when $E_* < 0$; this is the excluded endpoint resonance. With neither product vector, a nonzero dispersion vector cannot cancel because $E_* \notin \{-2, 0, 2\}$.

On the complement, a finite smooth partition supplies a real direction strictly separating all active vectors from zero on each chart. Write its ℓ singular factors as damped positive-half-line exponential integrals, and put $R = \sum_{i=1}^{\ell} \xi_i$. The exponent g has $\Re g \le 0$, $|e \cdot \nabla g| \ge cR$, and fixed angular jets $O(R)$. A fixed jth parameter derivative costs $O(R^j)$. Integrating by parts $\ell + j + 1$ times with $(e \cdot \nabla)/(e \cdot \nabla g)$ leaves radial majorant CR^{-2} for $R \ge 1$, including all coefficient and cutoff derivatives. The small-R part is bounded directly. Omitting both product factors gives the fixed-N, j onsite bound. □

Proposition 6.4 (Nonzero first amplitude)**.** *There is $L_* \neq 0$ such that*

$$T_{N_0}^{(k)}(\zeta_\varepsilon) = \begin{cases} 2L_*\varepsilon^{-1/2} + o(\varepsilon^{-1/2}), & N_0 \text{ even}, \\ 2L_*\varepsilon^{-1} + o(\varepsilon^{-1}), & N_0 \text{ odd}. \end{cases}$$

For even N_0 all lower derivatives are bounded. For odd N_0 the $(k-1)$st derivative is $O(1 + |\log \varepsilon|)$ and all smaller derivatives are bounded. The corresponding leading coefficient for C_{N_0} is $4L_*$.

Proof. Use fixed smooth weights depending only on the y angles around the two specified configurations, with disjoint supports. All x integrations remain unweighted, so the residues giving (6.1) apply to each chart exactly. Lemma 6.3 controls the full complementary double integral.

In the first chart write $y_i = re^{\mathrm{i}(-\alpha+u_i)}$, $v = \sum_i u_i$, $u_i = v/N + \eta_i$, $\sum_i \eta_i = 0$. The absolute real Jacobian in coordinates $(v, \eta_1, \dots, \eta_{N-1})$ is one. We use positive real product measures here; equivalently, the induced domain orientation compensates the determinant $(-1)^{N-1}$ of this coordinate change. Write $z_i = e^{-\mathrm{i}\phi_i}$. At the center,

$$b = \phi_u = \frac{\sin\alpha}{\sin\beta} > 0, \quad d = -\tfrac{1}{2}\phi_{uu} = \frac{H_{\alpha\beta}}{2\sin^3\beta} \neq 0, \qquad H_{\alpha\beta} = E_*(1 - \cos\alpha\cos\beta).$$

Also

$$\phi = \beta - \mathrm{i}A_0\varepsilon + bu - du^2 + O(\varepsilon^2 + \varepsilon|u| + |u|^3), \quad A_0 = \frac{2\sin\theta - c_0\sin\alpha}{\sin\beta} > c_0.$$

Replace the smooth mean cutoff by a fixed hard segment; its difference has separated product phases if the shape radius is sufficiently small. Move that segment downward in a fixed rectangle. The $Y = 1$ pole is at $v = -\mathrm{i}Nc_0\varepsilon$; downward motion increases the damping of Z. The vertical sides have a fixed phase gap and the bottom fixed damping, so all their fixed derivatives are bounded. The clockwise closure and $\partial_v(1 - Y) = -\mathrm{i}$ give the actual residue multiplier $+2\pi$. No nonholomorphic weight is moved through a complex contour.

At the residue the y_i lie on the unit circle and

$$1 - Z = Q\varepsilon - \mathrm{i}d\sum_i \eta_i^2 + O(\varepsilon^2 + \varepsilon|\eta| + |\eta|^3), \quad Q = \frac{2N\sin\theta}{\sin\beta} > 0, \tag{6.7}$$

with nonlinear bound $|1 - Z| \ge c(\varepsilon + \sum_i \eta_i^2)$. To see this bound, modulus damping is at least $c\varepsilon$; permutation symmetry cancels the linear shape term, and the real phase Hessian has a strict sign determined by $-H_{\alpha\beta}$. Its integral remainder controls the phase outside $\sum \eta_i^2 = O(\varepsilon^2)$, while damping controls the remaining region.

The exact numerator is $\Delta(\eta)^2 A(\zeta,\eta)$ with A analytic and nonzero at the center. Fixed parameter derivatives preserve the Vandermonde. Each pair contributes $-(\eta_i-\eta_j)^2/(4\sin^2\beta)$, and each one-body angle measure contributes $y_0 z_0/(2\pi \mathrm{i}\sin\beta)$. Since $y_0^N z_0^N = 1$, their combined phase is $(-1)^{N(N-1)/2}\mathrm{i}^{-N} = \mathrm{i}^{-N^2}$. After the actual mean residue the leading constrained value is

$$\frac{2\mathrm{i}^{-N^2}}{N!(2\pi)^{N-1}2^{N(N-1)}\sin^{N^2}\beta}\frac{\Delta(\eta)^2\,\mathrm{d}\eta_1\cdots\mathrm{d}\eta_{N-1}}{Q\varepsilon-\mathrm{i}d\sum_i\eta_i^2}.$$

After k derivatives and $\eta=\sqrt{\varepsilon}\,w$, only the top pole power contributes at scale $\varepsilon^{-1/2}$ for even N and ε^{-1} for odd N. The radial majorants have tails R^{-2} and R^{-3} respectively. Lower pole powers vanish after multiplication by the indicated singularity scale; splitting at $R=\sqrt{\varepsilon}$ gives the logarithm at the odd lower derivative. Dominated convergence applies to the shape integral after the mean residue.

For $a=(N^2-1)/2$, let

$$c_N=\frac{1}{\sqrt{N}}\int_{\sum\omega_i=0,\ |\omega|=1}\Delta(\omega)^2\,\mathrm{d}\omega>0.$$

The remaining radial period is

$$\int\frac{\Delta(w)^2\,\mathrm{d}w_1\cdots\mathrm{d}w_{N-1}}{(Q-\mathrm{i}d\sum_i w_i^2)^{k+1}}=\frac{c_N}{2}Q^{a-k-1}(-\mathrm{i}d)^{-a}B(a,k+1-a),$$

where the quadratic coefficient is continued from positive real part. Here $k+1-a$ equals $1/2$ or 1, and every factor is nonzero. For odd $N=p$, $a=k\equiv 0 \pmod 4$, so $d^k=|d|^k$ also on the negative-E_* family. The pole derivative is $Qe^{-\mathrm{i}\theta}$. In particular the complete one-chart coefficient is

$$L_*=\frac{\mathrm{i}^{-N^2}(-1)^k k!\,c_N B(a,k+1-a)\,2^a(2N\sin\theta)^{a-1}e^{-\mathrm{i}k\theta}}{N!(2\pi)^{N-1}2^{N(N-1)}(-\mathrm{i}H_{\alpha\beta})^a}. \tag{6.8}$$

Indeed the sine exponent is $-N^2-(a-1)+3a=0$. For even N, $H_{\alpha\beta}>0$ and the last power uses argument $-\pi/2$. For odd N, $a=k$ is divisible by four, so that power is $|H_{\alpha\beta}|^k>0$ for either sign of E_*. The factor i^{-N^2} equals one for even N and $-\mathrm{i}$ for odd N. Every remaining factor is symmetric in α,β. The two chart coefficients are equal as complex numbers, including the sign from the orientation, and hence add. The identity $C_N=f_N+2T_N$ and Lemma 6.3 give the factor four. □

6.3 A uniform boundary-tail theorem

The first singular coefficient is useful only if all higher particle orders are smaller at the same derivative order. The theorem below provides that comparison for the complete differentiated tail; its proof keeps the onsite terms and both signed kernels.

Negating (ζ,y,z) gives, directly from the forms and their measures,

$$T_N^{\sigma}(-\zeta)=T_N^{\sigma(-1)^N}(\zeta),\qquad f_N(-\zeta)=(-1)^N f_N(\zeta). \tag{6.9}$$

Conjugation follows from the real germ. Consequently it suffices to estimate either signed kernel at a selected point with $0<E_*<2$. Set $b_*=\arccos(E_*-1)$ and $\xi=e^{-\mathrm{i}b_*}$. Lemma 6.1, also applied after the reflection of the negative family, and Lemma 6.2 give the needed $|1\pm\xi^N|\ge e^{-AN}$.

Theorem 6.5 (The entire differentiated tail)**.** *For either parity, its selected first order N_0, and every fixed $0 \le j \le k$,*

$$\sum_{\substack{N>N_0\\ N \text{ of the selected parity}}} |T_N^{(j)}(\zeta_\varepsilon)| = o(\varepsilon^{-1/2}), \qquad \sum_{\substack{N>N_0\\ N \text{ of the selected parity}}} |C_N^{(j)}(\zeta_\varepsilon)| = o(\varepsilon^{-1/2}).$$

The estimates include all particle orders and both upper quadrants needed by the odd family.

We divide the proof into contour deformation, pair suppression, large-order estimates, and a separate estimate for the intermediate orders. The latter uses differentiation on the real integration domain and coarea bounds. The parameter disks are specified before the corresponding Cauchy estimates are applied. Fix the point and k first. Put

$$H = \log(1/\varepsilon), \qquad 0 < \alpha_0 < \frac{1}{2(k+2)}, \qquad \lambda_* = \varepsilon^{\alpha_0}.$$

All contour functions below are fixed smooth periodic functions, with bounded jets of every order required through k.

Choose m supported near the lower branch angle $-b_*$ and equal to one on a neighborhood of every lower branch-cutoff support. Choose a supported in $[\delta, \pi - \delta]$, equal to one on $[2\delta, \pi - 2\delta]$, and choose p supported in $(0, \pi)$ and equal to one on a neighborhood of $\operatorname{supp} a$. The upper inverse branch is inside $a = p = 1$, and the supports of m, p are disjoint. For a small fixed $\tau > 0$ define

$$P = \sum_i p(\theta_i), \quad h_i = -2\tau p(\theta_i) + \frac{\tau P}{2N} m(\theta_i), \quad y_i(\lambda, \theta) = re^{\mathrm{i}\theta_i + \lambda h_i}, \quad \psi = \prod_i (1 - a(\theta_i)).$$

Write $U_{\lambda,\zeta}$ for the pullback of the complete normalized form (6.1), including $1/N!$.

Lemma 6.6 (Exact weighted deformation)**.** *For every N and either σ,*

$$\begin{aligned} T_N^\sigma &= F_N + K_N + S_N, \\ F_N &= \int (1 - \psi) U_{1,\zeta}, \qquad K_N = \int \psi U_{0,\zeta}, \\ S_N &= -2\mathrm{i}\tau \int_0^1 \sum_q \int (\partial_{\theta_q} \psi) U_{\lambda,\zeta} \, \mathrm{d}\lambda. \end{aligned}$$

Every current term has one named upper index q with $p_q = 1$, $p_q' = m_q = m_q' = 0$; all other indices are in $\operatorname{supp}(1 - a)$.

Proof. The total radial shift satisfies $\sum h_i \le -3\tau P/2$, so $|Y| < 1$. Moreover,

$$\partial_{\log|y|} \Im W = -\cosh(\log|y|) \sin\theta.$$

Both inward upper motion and outward lower motion increase $\Im W$. All roots remain interior; the product and canceled-pair denominators remain nonzero throughout the closed homotopy.

After dividing row i of the angular Jacobian by y_i, its matrix is

$$M_{ij} = d_i \delta_{ij} + \frac{\lambda\tau}{2N} m_i p_j', \quad d_i = \mathrm{i} + \lambda \left(-2\tau p_i' + \frac{\tau P}{2N} m_i' \right).$$

Since $m_i p_i' = 0$, its determinant is $\prod_i d_i$. At a named current index its row and column are diagonal with $d_q = \mathrm{i}$; the homotopy replacement column multiplies the determinant by $2\mathrm{i}\tau$. Cartan's formula on the closed torus gives $\partial_\lambda \int \psi U_\lambda = -2\mathrm{i}\tau \sum_q \int (\partial_q \psi) U_\lambda$. Combine this with invariance of the unweighted closed integral. □

Support geometry and pair suppression. Here and below "branch" means the lower branch, never its excluded upper inverse. Choose nested branch cutoffs: an outer one supported in $|u| \le \delta_B$, equal to one on $|u| \le \delta_B/2$, and an inner one supported in $|u| \le 2r_0$, equal to one on $|u| \le r_0$, where $u = \theta + b_*$ and $r_0 = \delta_B/(2L^2)$ with L sufficiently large. All these supports lie in $m = 1$. Telescope the product of the outer cutoffs. A complementary term has an outer compact index with $|u_q| \ge \delta_B/2$, including all cutoff jets. Split the other coordinates by the inner cutoff and compact left/right labels. The latter can be made periodic by returning their transition inside the upper $a = 1$ plateau, where the full K and current weights vanish with all jets. Thus no cutoff differentiation introduces a new support type.

The partition has at most C^N labels. Its types are: all branch; a compact left root with a modulus gap; a separated compact/true-branch pair; and all compact right. A named current has no all-branch sector. If it has a branch index, choose the named upper index and a fixed branch index as the separated pair. The nested widths ensure that the branch slope dominates the compact slope by at least four; preliminary upper endpoint neighborhoods fix the latter bound before their final shrinking.

At $\varepsilon = \lambda = 0$, lower angles lie in $[-\pi, 0]$; their roots are real in $[z_0, 1]$ on $[-\pi, -b_*]$ and are $e^{-\mathrm{i}v}$, $0 \le v \le b_*$, on $[-b_*, 0]$, where $z_0 = E_* + 1 - \sqrt{(E_*+1)^2 - 1} > 0$. The identity

$$|1 - vw|^2 - |v - w|^2 = (1 - |v|^2)(1 - |w|^2) + 4\Im v \Im w \tag{6.10}$$

and its angular version imply $|P_{ij}| \le 1$. Equality is possible only for $y_i = 1$ paired with a left angle, or $y_i = -1$ paired with a right angle, with either ordering. Choose fixed overlapping groups

$$B = [-b_* - w, -b_* + w], \quad L = [-\pi, -b_* - \ell_L], \quad R = [-b_* + \ell_R, 0], \qquad 0 < \ell_L, \ell_R < w.$$

There is no equality configuration in a same-group square. For B^2 the angular Schur factor is at most $\sin w / \min(\sin(b_* - w), \sin(b_* + w)) < 1$ after shrinking w; the other groups have a strict root Schur bound. At the double branch the apparent singularity is removable: the inverse root coordinate is $y_\zeta(\phi) = G_\zeta(\phi^2)$ and

$$P_{ij} = (\phi_i - \phi_j)^2 A_\zeta(\phi_i, \phi_j), \qquad A_{\zeta_*}(0,0) = -\frac{1}{4\sin^2 b_*}, \tag{6.11}$$

with analytic bounded fixed jets. This follows by factoring both $y_i - y_j$ and $1 - e^{-\mathrm{i}(\phi_i + \phi_j)}$; the common $\phi_i + \phi_j$ factor cancels.

On the closed lower support, $1 - z_i z_j = 0$ forces the already resolved double root $z_i = z_j = 1$: the alternative $z_i = z_j = -1$ is excluded by $W_0 + 1 \ge E_* > 0$. Away from a smaller double-branch chart, this denominator therefore has a fixed positive minimum. The regular expression (6.11) and the original rational expression agree on their overlaps, and give a finite cover with bounded fixed jets. This also proves continuity at the common endpoints, where the canceled pair is zero. No reciprocal-root pole remains on that cover.

Choose a strict same-group bound $q < q_1 < q' < 1$ and cross-group slack $\eta > 0$ satisfying $\log(1+\eta) < -\log(q')/4$. Project the upper slivers $[0, 2\delta]$ and $[\pi - 2\delta, \pi]$ to their respective endpoints, assigning them to R, L. If $\bar\theta_i$ denotes this projection (and equals θ_i for lower angles), then uniformly in the entire independent box $0 \le \lambda, P/N \le 1$,

$$|y_i - e^{\mathrm{i}\bar\theta_i}| + |W_i - W_0(\bar\theta_i)| \le C(\varepsilon + \tau + \delta) + C|\zeta - \zeta_\varepsilon|.$$

Compact-root changes are Lipschitz, and branch changes are one-half Hölder by the square-root factorization and (6.11). The finite cover therefore gives a complete-pair change at most

$$C\sqrt{\varepsilon + \tau + \delta + |\zeta - \zeta_\varepsilon|}.$$

Choose $\delta, \tau, \varepsilon_0$ to make the radial-center change smaller than $\min(q_1 - q, \eta/2)$; reserve $q' - q_1$ and $\eta/2$ for the parameter disks. This proves both same-group and cross-group bounds on the slivers, uniformly in the actual coupled occupancy.

The named current index needs only bounded pairs. At its right limiting endpoint the root is $\xi = e^{-\mathrm{i}b_*}$, and for every limiting lower root z, $|1 - \xi z| = |e^{\mathrm{i}b_*} - z| \geq \sin b_* > 0$. At the left endpoint the root is $z_0 < 1$ and $|1 - z_0 z| \geq 1 - z_0$. These margins persist on the chosen neighborhoods. The order of choices is essential: the branch/upper-sliver limiting moduli are

$$\frac{\sin((b_* + a)/2)}{\sin((b_* - a)/2)} = 1 + a\cot(b_*/2) + O(a^2), \qquad \frac{\cos((b_* - a)/2)}{\cos((b_* + a)/2)} = 1 + a\tan(b_*/2) + O(a^2).$$

The excess over one is controlled by choosing the sliver widths after the compact-group margins and cross-group slack.

For $n = N$ lower variables in K, or $n = N - 1$ in a current, the number of same-group pairs is at least $n^2/6 - n/2$ and the number of cross pairs at most $n^2/3$. After deleting at most j factors, apply the bounds directly to the remaining product; no division by a possibly zero pair is allowed. The result is

$$\prod_{(i,l)\notin\mathcal{E}} |P_{il}| \leq C_j^N e^{-\kappa N^2}, \qquad |\mathcal{E}| \leq j, \tag{6.12}$$

for some fixed $\kappa > 0$. Named pairs cost only C^N.

The selected contour, including arbitrarily large N. Telescope $1 - \psi$ so that a selected index has $p_q = 1$ and hence $P \geq 1$. At $\lambda = 1$ it has a fixed interior root gap, and $\sum h_i \leq -3\tau P/2$ supplies a fixed Y gap. Every true branch has $\Im W \geq cP/N$, so its residue factor is bounded by $C\sqrt{N}$. These facts persist on a common disk $|\zeta - \zeta_\varepsilon| \leq c/N$. Branch roots remain in the lower unit half-disk; compact roots move by $O(1/N)$, and the fixed anchor absorbs their total inflation $e^{CN(c/N)}$ when c is small.

Let M be the number of compact indices. Divide their radial-center roots into a strict interior group and a compact lower-arc group. Same-group Schur factors are at most a fixed number below one. For a branch root w on the actual disk and a compact radial-center root z_0, the denominator $|1 - z_0 w|$ has a fixed positive lower bound, and

$$\frac{z - w}{1 - zw} - \frac{z_0 - w}{1 - z_0 w} = \frac{(z - z_0)(1 - w^2)}{(1 - zw)(1 - z_0 w)}.$$

Thus mixed factors are at most $1 + C/N$; only the compact root has been displaced in this comparison. It follows, even with the weaker $1 + C/\sqrt{N}$ budget, that the entire root Pfaffian product is at most $C^N e^{-aM^2}$.

Every y-pair involving a true branch is bounded: its reciprocal upper partner has inward shift -2τ, dominating the outward shift at most $\tau/2$. Compact pairs have the occupancy-independent majorant

$$\left|\frac{y_i - y_j}{1 - y_i y_j}\right| \leq \frac{C}{\varepsilon + \operatorname{dist}(\theta_i + \theta_j, 2\pi\mathbb{Z})}.$$

Indeed the sum of the two shifts is nonpositive at reciprocity and is $O(\operatorname{dist}(\theta_i + \theta_j, 2\pi\mathbb{Z}))$ nearby. A matched two-angle integral therefore costs $O(H + 1)$. Expand the angular Pfaffian into bounded minors and a matching on an even compact subset. Hadamard bounds the minor by $C^N N^{(N+1-2r)/4}$, the subset and matching count by N^{3r}, and the disjoint singular pair integrals by $(C(H + 1))^r$.

Including the one-body cost $C^N N^{N/2}$ and $N!$, the complete selected integral is bounded, for each assignment, by

$$C^N N^{-N/4+1/4}(CN^3(H+1))^M e^{-aM^2}.$$

The C^N assignments and named-index choices are absorbed in C^N. Completing the square in M, and applying Cauchy only to the complete integral with its fixed angular weights, gives

$$\sum_N |F_N^{(j)}(\zeta_\varepsilon)| \le C_j \exp(C_j \log^2(H+2)). \tag{6.13}$$

Here $e^{C_j \log^2(N+1)} N^{-N/4+j+1/4}$ is summable. The assignments are pointwise bounds for the full integral to which Cauchy's inequality is applied. The bound holds for all N and also with the product poles removed.

The original and protected current disks. On $|\zeta - \zeta_\varepsilon| \le c\varepsilon$, the undeformed imaginary margin and its monotone radial improvement keep all roots lower and interior. The two signed product gaps are at least $c'\varepsilon$. Near a true branch set $\mu = \lambda P/N$ and $u = \theta + b_*$. For $y = e^{v+\mathrm{i}\theta}$, $v = -c_0\varepsilon + \tau\mu/2$, the exact formulas

$$\Re W = \Re E - \cosh v \cos\theta, \qquad \Im W = \Im E - \sinh v \sin\theta$$

give an imaginary margin $c(\varepsilon+\mu)$ and real branch-center displacement $O(\varepsilon^2 + \mu^2 + |\zeta - \zeta_\varepsilon|)$. Splitting at $|u| = K(\varepsilon+\mu)$, first using the imaginary margin and then the linear real term, proves the occupancy-independent majorant

$$|R(y;\zeta)| \le C(|u| + \varepsilon)^{-1/2}. \tag{6.14}$$

Its integral is bounded. Compact factors and Jacobians cost C^N. Branch-coordinate motion on the disk is $O(\sqrt{\varepsilon})$; choosing ε_0 small retains the reserved pair slack, so (6.12) holds throughout the disk. Cauchy therefore gives

$$|K_N^{(j)}| + |J_{N,j}(\lambda)| \le C_j^N \varepsilon^{-j-2} e^{-\kappa N^2}, \qquad 0 \le \lambda \le 1, \tag{6.15}$$

where $J_{N,j}$ is the already differentiated, angularly integrated current. This bound will be used only for large N.

For a named current and $\lambda \ge \lambda_*$, $P \ge 1$ protects a larger disk $|\zeta - \zeta_\varepsilon| \le c\lambda_*/N^2$. Its branch margin is at least $c(\varepsilon + \lambda P/N)$. Choose its disk constant after τ so that the parameter displacement uses at most one quarter of this margin. Compact roots move by $O(\lambda_*/N^2)$; the named upper root has modulus at most $e^{-c\lambda}$ and absorbs the total compact inflation. Both product gaps are therefore at least $c'\lambda$. The same split proving (6.14), now including the disk displacement, gives the same integrable majorant. Pair-coordinate motion uses the remaining fixed slack in (6.12). Cauchy and then integration in λ yield

$$\sum_N \int_{\lambda_*}^1 |J_{N,j}(\lambda)| \, \mathrm{d}\lambda \le C_j \lambda_*^{-j-2} = C_j \varepsilon^{-\alpha_0(j+2)}. \tag{6.16}$$

The convergent sum $\sum_N C^N N^{2j} e^{-\kappa N^2}$ includes all orders. No large-current estimate is used to justify the original disk.

6.4 Real-domain differentiation in the intermediate window

Direct differentiation raises the powers of the product poles. Instead, a tangential correction to the parameter derivative keeps those kernels simple. The resulting derivatives fall on the numerator and cutoffs; the Vandermonde zeros compensate for the resulting collision losses. Vanishing puncture fluxes justify every integration by parts before the uniform estimates are taken.

The remaining orders satisfy $N_0 + 2 \le N < D\sqrt{H}$, with fixed D. For the small current $0 \le \lambda \le \lambda_*$, all finite compact jet errors $N^{C_j}(\varepsilon + \lambda_*)$ tend uniformly to zero in this window. We prove the required bound

$$|K_N^{(j)}| + \int_0^{\lambda_*} |J_{N,j}(\lambda)| \, \mathrm{d}\lambda \le e^{C_j N \log(N+1)} (H+N)^{C_j}. \tag{6.17}$$

Lemma 6.7 (Differentiation with simple kernels)**.** *Let w be a fixed smooth real angular weight, and let V be a parameter-dependent complex vector field. Put $\mathcal{D} = \partial_\zeta - \mathcal{L}_V$ on top forms. If all puncture fluxes vanish at every preceding stage, then*

$$\partial_\zeta^j \int w\Omega_\zeta = \int \mathcal{D}^j (w\Omega_\zeta).$$

If $(\partial_\zeta - V\cdot\nabla)Y = (\partial_\zeta - V\cdot\nabla)Z = 0$, both signed product kernels remain simple at every order.

Proof. The Lie derivative of a top form is a divergence. Integrate it on the fixed real domain, with punctures removed, and use the vanishing flux. Repeat on the entire preceding top form, including all derivatives of V, its divergence, and the previous coefficients. Complex coefficients are permitted by linearity of real integration by parts; no flow of a smooth cutoff into a complex domain is assumed. □

On an original branch chart set $\phi = \arccos W$ and $\chi_\zeta : u \mapsto \phi$. The branch one-body form is regular:

$$\frac{\mathrm{d}y}{2\pi \mathrm{i}} \frac{z}{\sinh\gamma} = -\frac{z}{\pi(1-y^{-2})} \, \mathrm{d}\phi.$$

Together with (6.11), the regular numerator is $\Delta(\phi)^2 A_\zeta(\phi)$ with analytic bounded fixed jets. Put $a_i = E'/W_{u_i}$, $b_i = \partial_{u_i}\phi_i$, $A = \sum_i a_i$. For a selected pair p, q use

$$V_i = a_i \ (i \ne p, q), \qquad V_p = a_p + \frac{Ab_q}{b_p - b_q}, \qquad V_q = a_q - \frac{Ab_p}{b_p - b_q}.$$

Then $\sum V_i = 0$ and $\sum b_i V_i = \partial_\zeta \sum \phi_i$. The residual regular-coordinate field $B = \partial_\zeta \chi_\zeta - (\chi_\zeta)_* V$ has only

$$B_p = -\frac{Ab_pb_q}{b_p - b_q}, \qquad B_q = -B_p.$$

Because $b(\phi) = B_\zeta(\phi^2)/\phi$ with B_ζ analytic and $B_\zeta(0) \ne 0$, one has the exact analytic factorization

$$\phi_q B_\zeta(\phi_p^2) - \phi_p B_\zeta(\phi_q^2) = (\phi_q - \phi_p)\{B_\zeta(0) + O(|\phi_p\phi_q|)\}.$$

The expression in braces is bounded away from zero on a smaller fixed chart. Thus $b_pb_q/(b_p - b_q)$ is an analytic bounded coefficient divided by $\phi_q - \phi_p$, with bounded required jets. All fixed coefficient jets have polynomial N-cost apart from this explicit selected difference pole. For smooth real w the exact pullback identity is

$$(\partial_\zeta - \mathcal{L}_V)(w\chi_\zeta^*\Omega) = w\chi_\zeta^*(\partial_\zeta + \mathcal{L}_B)\Omega - (V\cdot\nabla_u w)\chi_\zeta^*\Omega. \tag{6.18}$$

It follows by differentiating the Jacobian determinant and subtracting the two divergence formulas. The real cutoff is never holomorphically extended.

After j operations, each term has a derivative $\partial_u^\mu w$, a numerator derivative $\partial_\zeta^\ell \partial_\phi^\nu(\Delta^2 A)$, the unchanged simple kernels, and a coefficient of selected pole order m satisfying

$$m + |\mu| + |\nu| \le 2j, \qquad \ell + |\mu| + |\nu| \le j. \tag{6.19}$$

The filtration follows by checking each product-rule term. A fixed-ϕ parameter derivative of a coefficient adds no difference pole; $B \cdot \nabla$ on a coefficient, or multiplication by $\operatorname{div} B$, adds at most two; a numerator-coordinate derivative adds one derivative and at most one pole; a numerator-parameter derivative adds only one derivative; and a cutoff derivative adds one cutoff derivative and at most one pole. These are all product-rule terms in (6.18). There are $N^{O(j)}$ descendants. At most j complete pair factors are changed in each descendant, so (6.12) survives on the others. The recurrence retains the differentiated numerator, since division by the original Vandermonde would introduce poles at unselected collisions.

To select pairs smoothly, fix an integer $M > 2k + 2$ and use

$$w_{pq} = \frac{(u_p - u_q)^{2M}}{\sum_{i<l}(u_i - u_l)^{2M}}, \quad d_u = \max_{i<l} |u_i - u_l|.$$

Off full equality these sum to one and obey

$$|\partial_u^\mu w_{pq}| \le C_k N^{C_k} \frac{|u_p - u_q|^{2M-|\mu|}}{d_u^{2M}}.$$

The analytic inverse branch is $u = U_\zeta(\phi^2)$, with its fixed contour constants included. Its bounded derivative gives

$$|u_p - u_q| \le C|\phi_p^2 - \phi_q^2| \le C'|\phi_p - \phi_q|.$$

The rational weights therefore absorb the selected poles, losing at most $2j$ powers of shape radius. A further power is reserved for a transition or flux estimate. At fixed ε, N, a selected-diagonal tube of radius h has flux $O(h^{2M+1-2j})$ on a separated support. At full equality, the shape surface measure gives flux exponent

$$N(N-1) - 2j - 1 + N - 2 = N^2 - 2j - 3 > 0 \quad (N \ge N_0 + 2,\ j \le k).$$

Thus all puncture fluxes vanish, in this order, at every preceding Lie stage. These checks justify the identities before taking boundary-uniform bounds.

Branch geometry used in coarea. For the original contour, near $u = 0$,

$$1 - W = a_* u - \mathrm{i} b_*' \varepsilon + O(u^2 + \varepsilon|u| + \varepsilon^2), \quad a_* = \sin b_* > 0, \quad b_*' = 2\sin\theta - c_0 a_* > 0.$$

The exact factorization $\phi = \sqrt{2(1-W)}\, h(1-W)$, $h(0) = 1$, gives

$$|\phi'| \asymp (|u| + \varepsilon)^{-1/2}, \quad \int |\phi'| \, \mathrm{d}u \le C, \quad -\Im\phi \ge c\sqrt{|u| + \varepsilon} \quad (u \le 0).$$

For $u \ge 0$, $x' = \partial_u \Re\phi \asymp (u + \varepsilon)^{-1/2}$, and $-x'' \ge cu^{-3/2}$ for $u \ge C_0\varepsilon$. If $M_1/\varepsilon \to \infty$ and $0 \le m_1 \le M_1/2$, then

$$x'(m_1) - x'(M_1) \ge c(m_1 + \varepsilon)^{-1/2}, \quad \frac{|\phi'(m_1)\phi'(M_1)|}{x'(m_1) - x'(M_1)} \le C M_1^{-1/2}.$$

For $M_1/2 < m_1 < M_1$, with $m_1 \gg \varepsilon$, the last ratio is at most $C\sqrt{M_1}/(M_1 - m_1)$. These follow from the leading square-root derivative, using its $u^{-3/2}$ curvature in the comparable case. On a true branch plateau of the small current the same factorization holds with scale $\varepsilon + \lambda P/N$. In particular $\Re\phi_u \geq (2/3)|\phi_u|$ for $u \geq 0$, and

$$\frac{|\phi_u'|}{|1 - \sigma Z|} \leq \frac{C}{|u| + \varepsilon + \lambda P/N} \quad (u < 0). \tag{6.20}$$

The possible real branch-center displacement is quadratic in this scale; it is retained in deriving these bounds. A plateau variation fixes P.

All-branch microcore. Choose $b_N = c_* e^{-(A+4)N}/N$ and localize by $\prod_i \chi(u_i/b_N)$, with $\chi = 1$ on $[-1/2, 1/2]$ and supported in $[-1, 1]$. Uniformly for $N < D\sqrt{H}$, $N\varepsilon e^{AN} \to 0$ and $\varepsilon/b_N^m \to 0$ for each fixed m. Therefore $|1 - \sigma Y| \geq \frac{1}{2}e^{-AN}$ on the core and all cutoff jets. The one-phase field $V_i = a_i$ freezes every regular ϕ_i, hence Z, and preserves the Vandermonde polynomial. Its jth derivative has only $e^{O_j(N)}$ cutoff and Y-kernel losses and $e^{O_j(N^2)}$ analytic product cost. The branch image has radius $C\sqrt{b_N}$ in a fixed cone; choosing c_* small makes $N\sqrt{b_N}$ small and $|1 - \sigma Z| \geq c\sum_i |\phi_i|$ (a weaker bound also valid for $\sigma = -1$). The Laplace identity for the reciprocal of this sum and

$$\int_{\text{core}} e^{-t|\phi|} |\,\mathrm{d}\phi| \leq C \min(\sqrt{b_N}, t^{-1})$$

bound the core derivative by

$$e^{C_j N^2} b_N^{(N^2-1)/2}. \tag{6.21}$$

Its logarithm is $-(A+4)N^3/2 + O_j(N^2 \log(N+1))$, and the series is bounded independently of ε.

All-branch complement. Telescope the complement of the microcore. A named anchor satisfies $|u_a| \geq b_N/2$ on every cutoff jet. Put $\bar{u} = N^{-1}\sum u_i$, $\rho^2 = \sum(u_i - \bar{u})^2$ and split smoothly at $\rho_N = e^{-BN}$, where B is fixed large enough that $2\rho_N \leq b_N/8$ for every $N \geq N_0 + 2$.

On $\rho \geq \rho_N$, maximal separation is at least $\rho_N/\sqrt{N}$. The Lie losses and cutoff jets cost $e^{O_j(N\log(N+1))}$ and unchanged pairs retain quadratic suppression. If all coordinates are nonnegative, select the actual minimum m_1 and maximum M_1 only in this final absolute integral. For fixed other variables the phase map

$$(m_1, M_1) \longmapsto (m_1 + M_1 + \text{constant},\ x(m_1) + x(M_1) + \text{constant})$$

has positive Jacobian $x'(m_1) - x'(M_1)$ and multiplicity one on each extreme-index cell. The preceding branch estimates absorb the two branch measures at cost $e^{O(N)}N^{O(1)}$; here $M_1 \geq b_N/2$, and in the comparable case $M_1 - m_1 \geq \rho_N/\sqrt{N}$. The two unwrapped phase intervals have bounded length for fixed other variables.

If some negative coordinate has magnitude at least $\delta_N = e^{-C_1 N} \leq b_N/16$, then Z has gap $e^{-O(N)}$; integrate the remaining Y kernel in the named anchor, whose measure costs at most $e^{O(N)}$. If every negative coordinate has magnitude below δ_N, the anchor is positive; use one negative coordinate with (6.20) and use the anchor for the Y kernel. Both integrations cost at most $e^{O(N)}(H + N)$.

On $\rho \leq 2\rho_N$, the anchor gives $|\bar{u}| \geq 3b_N/8$, so all coordinates have the same sign and $|u_i| \geq b_N/4$. The branch measures and the conversion of the Vandermonde to the angular shape radius cost at most

$$C^{N^2} b_N^{-N^2/2} = \exp\left(\tfrac{A+4}{2}N^3 + \tfrac{1}{2}N^2 \log N + O(N^2)\right).$$

The remaining shape power after Lie losses and coarea is

$$L_N = N^2 - 2j - 4 \geq \begin{cases} 4N_0 + 2, & N_0 \text{ even}, \\ 4N_0 + 1, & N_0 \text{ odd}. \end{cases}$$

For negative mean the Z gap is $e^{-O(N)}$ and a single mean phase suffices. For positive mean the phase map $(\bar{u}, \rho) \mapsto (N\bar{u} + \text{constant}, \sum_i x(u_i))$ has Jacobian at least $cN\rho$, by strict concavity and the zero first shape derivative. It has multiplicity one on each shape ray. Bound ρ^{L_N} by $(2\rho_N)^{L_N}$ before integrating the phase kernels. Both unwrapped phase lengths are $O(N)$, giving an explicit factor $CN^2(H+N)^2$. The logarithm of the total bound is

$$\left(\tfrac{A+4}{2} - B\right) N^3 + \tfrac{1}{2} N^2 \log N + O_j(N^2) + B(2j+4)N,$$

apart from that phase factor. Choose $B > A + 4$ after fixing the anchor-neighborhood constants. This is summable; finitely many small orders have fixed, possibly large, constants. The same positive power already proved the required full-equality puncture fluxes vanish.

How the phase integrations are estimated. For the simple-kernel integrations just used, the elementary inequality

$$|1 - e^{-a+\mathrm{i}t}|^{-1} \leq C\big(\min(1, a) + \operatorname{dist}(t, 2\pi\mathbb{Z})\big)^{-1}$$

gives $C(1+\ell)(1+|\log a|)$ on an interval of length ℓ. For two phases, divide out the specified real Jacobian, enlarge to the containing rectangle, and apply this estimate twice. The interval lengths account for periodic wrapping, and the unwrapped multiplicity comes from the coarea map. The phase replaces the coarea radius in this integral.

Mixed and compact sectors. Use regular ϕ_i on true branch indices J, and original angles on compact indices. Let q be the named compact anchor, choose $j_0 \in J$, and put

$$A = \sum_{i \in J} a_i, \quad D_0 = \sum_{i \notin J} \partial_\zeta \phi_i, \quad X_0 = \frac{D_0 + b_q A}{b_{j_0} - b_q}.$$

Set $V_i = a_i$ for $i \in J \setminus \{j_0\}$, $V_{j_0} = a_{j_0} + X_0$, $V_q = -A - X_0$, and all others zero. This freezes Y, Z and every unselected branch phase. The sole residual branch velocity is $-(D_0 + b_q A)/(1 - b_q/b_{j_0})$. The nested support has $|b_q/b_{j_0}| \leq 1/4$, so this coefficient and all fixed jets have polynomial N-cost in regular coordinates; there is no selected-difference pole. On current supports the chosen indices are on true plateaus, giving $V \cdot \nabla P = V \cdot \nabla \sum m_i = 0$ identically, including parameter derivatives. The hybrid Jacobian is triangular; its branch measure is precisely the regular one displayed above, and its compact factor retains the full $\prod_i d_i$ determinant.

For positive branch coordinate, the phase Jacobian in the compact anchor and selected branch is $\Re(b_{j_0} - b_q) \geq (2/3 - 1/4)|b_{j_0}|$; its fixed-first-phase map is injective and absorbs the selected branch measure. If a branch coordinate is negative, use (6.20) instead. With a compact left root, Z has a fixed gap: take $V_i = a_i$ on branches and $V_q = -A$, freezing Y and each branch root, and differentiate the separated Z kernel at polynomial cost. These alternatives prove (6.17) for the mixed and left sectors using (6.12) and the simple phase estimates.

On an all compact right interval put $g(\theta) = \arccos(E_* - \cos\theta)$. Direct differentiation gives

$$g''(\theta) = -\frac{E_*(1 - \cos\theta\cos g)}{\sin^3 g} \leq -c_R < 0.$$

The coupled-contour finite jets differ from their limiting values by $N^{O(j)}(\varepsilon+\lambda)$, uniformly small in this window. Thus $G=\Re\sum_i\phi_i$ retains a strictly negative Hessian. Let $f_i=\partial_{\theta_i}\log Y$, $g_i=\partial_{\theta_i}\sum\phi_i$, and $Q_0=\partial_\zeta\sum\phi_i$. By permutation symmetry and smooth division,

$$f_pg_q-f_qg_p=(\theta_p-\theta_q)c_{pq},\qquad |c_{pq}|\ge c_R/2.$$

The field $V_p=-Q_0f_q/(f_pg_q-f_qg_p)$, $V_q=Q_0f_p/(f_pg_q-f_qg_p)$ freezes both kernels. The same five-case induction and rational weights lose at most two shape powers per Lie order. Coupled occupancy is included: keep each complete pair as a scalar function of all angles. A derivation changes at most one new pair per product-rule step, even when its occupancy derivative moves many roots. Thus each descendant changes at most j pairs, has $N^{O(j)}$ terms, and retains (6.12) on the others.

For maximal separation at least N^{-K_0} all selected losses are polynomial. The corresponding real coarea assertion also holds for the genuinely coupled contour. Fix the other angles and the mean of an ordered extreme pair. By permutation symmetry $G(\dots,m+d,\dots,m-d,\dots)$ is even in d, and its second derivative is at most $-2c_R'$ for a fixed $c_R'>0$. Integrating from $d=0$ gives

$$|\partial_pG-\partial_qG|\ge c_R'|\theta_p-\theta_q|.$$

Consequently the phase map $(\theta_p,\theta_q)\mapsto(\theta_p+\theta_q,G)$ has multiplicity one on each ordered extreme-index cell and Jacobian at least c_R' times the maximal separation. Its coarea cost is polynomial in N; unwrapped phase lengths $O(N)$ give a further $N^{O(1)}(H+\log N)^2$ factor by the simple-kernel estimates. Below that scale use $\theta_i=\mu+\rho\omega_i$, $\sum\omega_i=0$, $\sum\omega_i^2=1$. In this range $\rho\le\sqrt{N}\,N^{-K_0}$. The phase map $(\mu,\rho)\mapsto(N\mu,G)$ has Jacobian at least $c_RN\rho/2$ and multiplicity one on each ray. The numerator costs $C^{N^2}\rho^{N(N-1)}$. After the same conservative losses its remaining coarea power is $L_N>0$ above. A fixed sufficiently large $K_0>1$ absorbs C^{N^2} and the polynomial field costs. The phase rectangle has side lengths $O(N)$ and attenuation floors $cN\varepsilon$, so its integral costs $CN^2(H+\log N)^2$. Rational-weight and full-equality fluxes vanish as already checked. This proves (6.17) in the last support type.

Onsite terms and completion of the tail. For f_N repeat the exact weighted deformation with both product poles deleted. The selected estimate (6.13) remains valid. For the other pieces use only the one-phase branch field $V_i=(\partial_\zeta\phi_i)/(\partial_{\theta_i}\phi_i)$ and zero velocity on compact indices. It freezes each branch phase, has bounded analytic coefficients, and preserves occupancy on the branch plateaus. There are no selected poles or punctures. Every fixed derivative changes at most j pairs; regular branch lengths are uniformly integrable by (6.14). Thus the remaining pieces are bounded by $C_j^NN^{C_j}e^{-\kappa N^2}$ for every N and $0\le\lambda\le1$. Consequently

$$\sum_N|f_N^{(j)}(\zeta_\varepsilon)|\le C_j\exp(C_j\log^2(H+2))=o(\varepsilon^{-\delta})\quad\text{for every }\delta>0.\tag{6.22}$$

Differentiate the locally normally convergent series first. Only afterwards split its already differentiated terms and current at $N=D\sqrt{H}$ and λ_*. Choose a common κ in the pair estimates and then D with $\kappa D^2>k+2$. The selected sum is $e^{O_j(\log^2(H+2))}$. The large current is $O_j(e^{\alpha_0(j+2)H})=o(e^{H/2})$. The entire remaining high-order tail from (6.15) is at most

$$\exp((j+2-\kappa D^2)H+O_j(\sqrt{H}))=o(e^{H/2}).$$

Summing (6.17) over all intermediate orders costs $\exp(O_j(\sqrt{H}\log(H+2)))=o(e^{H/2})$. This gives the tail bound for signed T_N^σ at positive E_*. Equation (6.9) and conjugation give the other odd family. The full-site bound follows from (6.22) and $C_N=f_N+2T_N$.

Order of constants and quantifiers. The point, N_0, k, radial margin and all required derivative orders are fixed first. Next fix branch and compact margins and the nested slope separation; then the same-group pair bound, its cross slack, and only then the endpoint slivers and τ. Choose disk constants after these data, preserving the branch and named-anchor gaps. Next fix α_0, D, M, K_0 and the algebraic separation constant A. Choose the microcore prefactor and determine its anchor-neighborhood costs before choosing B, C_1 large enough for the inclusions at every $N \geq N_0 + 2$ and the negative cubic coefficient. Finally choose one ε_0 small enough for all the preceding estimates. The elementary limits $\varepsilon^r e^{CN} N^m \to 0$ for $N \leq D\sqrt{H}$ and fixed $r > 0, C, m$ make this single choice uniform in that window. Finitely many small-order constants may be large but remain independent of ε. The particle and current windows are imposed after differentiation.

6.5 Dense local obstructions and the exact physical identities

We now restore the physical normalization. The first singular coefficient and the complete tail bound combine before any density argument is used. Exact coefficient identities then relate the isotropic functions to the equal-coupling restriction of the joint functions.

The full-site form-factor identification gives, on the respective real physical intervals,

$$\mathcal{X}_L(s) = (1 - s^{-4})^{1/4} \sum_{N \geq 2,\ N \text{ even}} C_N(s), \qquad s > 1, \tag{6.23}$$

$$\mathcal{X}_H(s) = \frac{(1 - s^4)^{1/4}}{s} \sum_{N \geq 1,\ N \text{ odd}} C_N(1/s), \qquad 0 < s < 1. \tag{6.24}$$

The low-temperature formula has the plus-phase subtraction already performed: its omitted zero-particle term is M_+^2, and its onsite contribution is $1 - M_+^2$. Equation (6.4) therefore also gives the exact alternative $\mathcal{X}_L = 1 - M_+^2 + 2M_+^2 \sum_{N \geq 2,\ N \text{ even}} T_N$. The high-temperature onsite sum is one, and $C_1 = 1/(E - 2)$ checks the normalization. The branches in (6.23)–(6.24) are positive on their stated real intervals.

Proposition 6.8 (Dense obstructions for the physical isotropic sums)**.** *The full normalized even exterior sum in* (6.23) *and the full normalized odd interior sum in* (6.24) *have no local holomorphic extension at a dense set of points of $|s| = 1$. At their selected points the nonzero leading derivatives are those of Proposition 6.4, multiplied by the physical nonvanishing factors. These assertions use the full-tail bounds of Theorem 6.5.*

Proof. At every selected point, Proposition 6.4, the bounded lower orders, and Theorem 6.5 give a nonzero leading derivative of the complete physical sum. All lower derivatives are smaller than that leading scale, so multiplication by its local analytic nonzero phase prefactor preserves it by the product rule. A holomorphic extension through such a point would bound that fixed finite derivative on the short radial segment, a contradiction. The selected points are dense on the required arcs. For the even sum, the contour transformation $(s, x, y) \mapsto (-s, -x, -y)$ gives evenness, and conjugation completes all four quadrants. For the odd sum both signs were treated explicitly; conjugation completes a dense subset of the circle. The inversion $s = 1/\zeta$ is locally biholomorphic at each unit-circle point, so the odd conclusion concerns the physical interior germ. This proves the stated local obstructions. Their passage to all boundary points and their role in the exact slice and isotropic corollaries are established in Section 7. □

The high-temperature value at $s = 0$ is the removable value one. This can also be checked without the joint representation: uniformly in the real angular variables, $z = s/2 + O(s^2)$, each

one-body factor is $O(s)$ and each squared pair is $O(s^2)$, giving $C_N(1/s) = O(s^{N^2})$ with a bound $C^{N^2}|s|^{N^2}/N!$ for sufficiently small $|s|$. The $N = 1$ coefficient is $s/(1-s)^2$, and every higher odd order vanishes after division by s. The exterior normalized low-temperature germ is single valued; the energy-field susceptibility is interpreted on the physical continuation of $\beta_{\mathrm{phys}} = \operatorname{arsinh}(s)/(2J)$. At every selected nonaxis point this factor is analytic and nonzero, so the same dense obstructions hold for $\chi = \beta_{\mathrm{phys}}\mathcal{X}$. On the exterior the inverse temperature is read on the physical cover.

Finally put $s = \sinh z$ on the equal-coupling restriction. The central cell

$$|\Im z| < \pi/2, \qquad |\Re z| < \operatorname{arsinh}(\cos \Im z)$$

maps biholomorphically onto $|s| < 1$. The right cell

$$\Re z > \operatorname{arsinh}(|\cos \Im z|)$$

is the universal cover of $|s| > 1$ under the same map, with deck translations $z \mapsto z + 2\pi\mathrm{i}$. One direct verification writes $w = e^z$ and solves $w - w^{-1} = 2s$: for $|s| > 1$ exactly one root has $|w| > 1$, since a unit root would give $s \in \mathrm{i}[-1, 1]$. Exponentiation supplies the stated covering. At the selected nonramified boundary points the local isotropic obstructions pull back directly. These identities fix the entire even/odd sums and the physical factor $\beta = z/(2J)$ used in the proof of the slice theorem and its isotropic descendants below.

7 Proofs of the joint, slice and isotropic theorems

7.1 The complete joint obstruction

Proof of Theorem 1.1. Theorem 2.1 determines the spectral set at every coupling, including collinear and zero-real-part degeneracies. Theorem 2.2 identifies the physical components and their contractibility and p-bounds. Proposition 2.18 gives their plurisubharmonic exhaustion. The physical form-factor and coefficient identities with Proposition 3.1 give joint holomorphy and normal convergence of the physical sums.

Take a regular face point whose free coefficient is $b = \cos(2\pi a/\ell)$, with ℓ an odd prime and $0 < a < \ell/2$. By Proposition 2.10 the variables (C, B), after exchanging transfer axes if needed, are independent holomorphic local coordinates, $\Im B \neq 0$, and the physical domain is the full complementary side of that edge. At such a point $S, T \neq 0$, and the edge identity (2.18) implies $p^2 \neq 1$. Thus the locally chosen physical prefactor is analytic and nonzero.

At a positive endpoint, restrict to any holomorphic transversal with nonramified edge coordinate $u = C - B - b$, and write its anisotropy as $B(u)$. Conjugating the data when needed gives $\Im B(0) > 0$, and the physical side contains $u = \mathrm{i}\varepsilon$. Proposition 4.3 proves a nonzero singular derivative of the entire odd or even sum, including the complete differentiated higher tail and physical prefactor. The same proposition handles a negative endpoint for the even sum by the exact angular shift and, if necessary, conjugation.

For a negative endpoint of the odd sum use

$$t = \frac{C + B - b}{B}.$$

This is also nonramified on the stipulated transversal, because $B(0) \neq 0$. In the lower-half-root orientation the anisotropy $B(t)$ has $\Im B(0) < 0$, and the local physical side is $\Im(B(t)t) > 0$. Theorem 5.1 gives all finite boundary jets of the complete odd sum and a selected sequence whose coefficient roots diverge. Its exact full-sum asymptotic (5.84) excludes cancellation by every other

particle index and winding. Division by the nonvanishing physical prefactor proves the same nonextension for the physical response. The opposite orientation follows by conjugation. Sections 4 and 5 give the estimates uniformly on the indicated compact complex-parameter families.

In particular choose a constant-B transversal. If the joint physical function extended to a neighborhood of the selected point, its restriction to that disk would give the forbidden one-variable extension. Thus the transversal obstruction gives ambient nonextension at each selected point. The selected points are dense on regular faces by Proposition 2.10, and regular faces are dense in the entire physical boundaries by Proposition 2.9. Corollary 2.16 then excludes componentwise extension at every boundary point, including corners, collinear points and vanishing transfer-coordinate charts. On the collinear equal-coupling restriction the local obstructions are those of Proposition 6.8.

Finally consider analytic continuation in an unramified Riemann domain over $\mathbb{C}^2$ from one of these physical germs. As long as the projected continuation path remains in its domain, uniqueness of analytic continuation makes every continued germ the germ of the single-valued function already constructed there. If a path first leaves that domain, its first exit is a boundary point. A local biholomorphic chart of the Riemann domain at the continued point would yield a holomorphic function on a base neighborhood agreeing with the physical function on an accumulating domain-side component. The proved componentwise obstruction forbids this. Compactness of the finite path interval guarantees a finite first exit whenever the path reaches outside. The obstruction therefore holds on every unramified continuation reached from the physical germ. □

7.2 All physical ratios, including the collinear one

Lemma 7.1 (Complex neighborhoods of the coupling ratio)**.** *At a regular selected point on a positive-ratio slice with nonzero edge derivative, the transverse representation is holomorphic in the complex ratio λ nearby. The boundary estimates hold uniformly on common compact parameter polydisks.*

Proof. The boundary temperature coordinate z_0 is nonzero. Thus $(z, \lambda) \mapsto (z, w = \lambda z)$ has determinant z_0 and is a local biholomorphism. Its composition with the regular (C, B) coordinate map is holomorphic. If the positive-edge parameter is $u = C - B - b$, its derivative in z is nonzero by hypothesis; the holomorphic implicit-function theorem gives $z = z(u, \lambda)$ and hence $B = B(u, \lambda)$ on a product polydisk. At a negative edge use $t = (C + B - b)/B$ in the same way. Shrinking the polydisk preserves the strict sign of $\Im B$, the nonvanishing physical prefactor and the other strict face conditions. Cauchy estimates on a slightly larger polydisk give the common holomorphic bounds required in Proposition 4.3 and Theorem 5.1. For a negative edge the selected integers may depend on the complex ratio. The equal-coupling boundary is treated by the collinear estimates, and joint holomorphy on interior compact sets also applies in complex-ratio neighborhoods at $\lambda = 1$. □

Proof of Theorem 1.2. The exact open-set equalities are Theorem 2.6. The physical sums of Proposition 3.1 restrict holomorphically to these pullbacks. A pullback boundary point maps to the corresponding joint boundary by the open-set equality and continuity.

Fix $\lambda \neq 1$. Proposition 2.12 gives a dense set on this slice at which the edge map has nonzero derivative and the free phase is a chosen prime-grid value. At each such point the line is a holomorphic curve $B = B(u)$ or $B = B(t)$ of the type used in Theorem 1.1. Apply Proposition 4.3 or Theorem 5.1 to that varying anisotropy. The restricted full sum is nonextendable there. The one-variable collar in Lemma 2.14 and Corollary 2.16 extend the obstruction to every boundary point.

Now let $\lambda = 1$. The exact coefficient and normalization identities in (6.23)–(6.24) identify the restriction with the full even and odd integrals of Section 6; its local obstructions were proved in Proposition 6.8. Lemma 2.7 maps the two slice domains to the exterior and disk, respectively. At every boundary point with $\cosh\zeta \neq 0$ this is a local biholomorphism. The selected obstructing points pull back densely on the nonramified boundary arcs. The remaining points with $\cosh\zeta = 0$ are limits of those arcs. The same local collar therefore proves nonextension everywhere on both slice boundaries.

Section 3 identifies the energy-field response $\chi = \beta\mathcal{X}$, with the plus phase fixed in LT. The complete coefficient sum includes the origin and both axes and subtracts the zero-particle M^2 in LT. The entire factor $\beta = \zeta/(2J_x)$ is nonzero on both boundaries, since zero lies in the HT interior and strictly outside the LT closure. It therefore preserves local boundary extendability. □

7.3 Derivation of the complete isotropic theorems from the slice theorem

Proof of Corollary 1.3. Apply the now proved Theorem 1.2 with $\lambda = 1$. The identities $S = T = s$, $p = s^2$, $B = 1$, and

$$\mathcal{D}(\zeta_1, \eta_1; z, z) = s\left[s + s^{-1} - \tfrac{1}{2}(\zeta_1 + \zeta_1^{-1} + \eta_1 + \eta_1^{-1})\right]$$

identify the parent coefficients term by term with the full isotropic coefficients. On the real LT interval $\sqrt{p} = s$, so each factor $\sqrt{p}/\mathcal{D}$ is exactly its isotropic counterpart. The even prefactor becomes $(1-s^{-4})^{1/4}$. On the real HT interval the dispersion is $s+s^{-1}-\cos q$ and the odd prefactor is $(1-s^4)^{1/4}/s$, giving the exterior auxiliary variable $1/s$. Normal convergence and the identity theorem extend these identities to their physical complex domains.

For HT, the biholomorphism in Lemma 2.7 transports the restricted holomorphic function to the disk. The removable value at zero is one, independently checked at the end of Section 6. For LT, the normalized termwise identities show invariance under every deck translation $z \mapsto z + 2\pi\mathrm{i}k$; the full normalized function therefore descends from $\Omega_{L,1}$ to a single-valued holomorphic function on $|s| > 1$. It is the positive real LT germ and has the stated connected onsite contribution.

Suppose one of these descended functions extended through a point s_0 of the unit circle. Choose a boundary preimage z_0 on its physical slice. Composing the proposed extension with the entire map sinh would extend the slice function near z_0, in agreement on the physical side. This contradicts Theorem 1.2. Composition with sinh also applies at its ramified boundary values $s_0 = \pm\mathrm{i}$. Thus the unit circle is a natural boundary for both normalized germs.

Finally $\beta = z/(2J)$ on the prescribed physical temperature continuation yields the energy-field claims. In the disk it is the branch $\operatorname{arsinh}(s)/(2J)$. In the exterior it is interpreted on the proved covering. At all selected nonaxis points it is analytic and nonzero, so the local singular derivative remains nonzero after the product rule; density gives all remaining boundary points. □

8 Decoupled restriction and an interior spectral point

The decoupled restriction $K_y = 0$ is entire by Remark 3.2, including across its intersections with the joint frontier. The fixed-ratio theorem uses the transversals on each slice and the collinear argument at equal coupling.

The inclusion $\{|p| = 1\} \subset \Sigma$ is strict. At

$$z = \mathrm{i}\pi/2, \qquad w = \operatorname{arsinh}(1/2)$$

one has $A = 0, S = \mathrm{i}, T = 1/2$, so $a = b = 0$ represents a spectral point, while $|p| = 1/2$. In fact $F_1 = -1/4, F_2 = 1$, so this point lies in the interior of Σ.

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